# ZOO501 – MIDTERM MERGED PPT (LECTURE 01 TO 72) Developmental Biology


Prepared By:

**All About Biotech**

AARISH ALI & TEAM (+96569002641)


Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

# Introduction to Developmental Biology

# Introduction to Developmental Biology

All multicellular organisms arise by a slow process of progressive change called ***development.*** Development is a process by which a multicellular organism arises, initially from a single cell- the fertilized egg, or **zygote,** which divides mitotically to produce all the cells of the body.

# Introduction to Developmental Biology

The study of animal development has traditionally been called **embryology**, from that stage of an organism that exists between fertilization and birth.

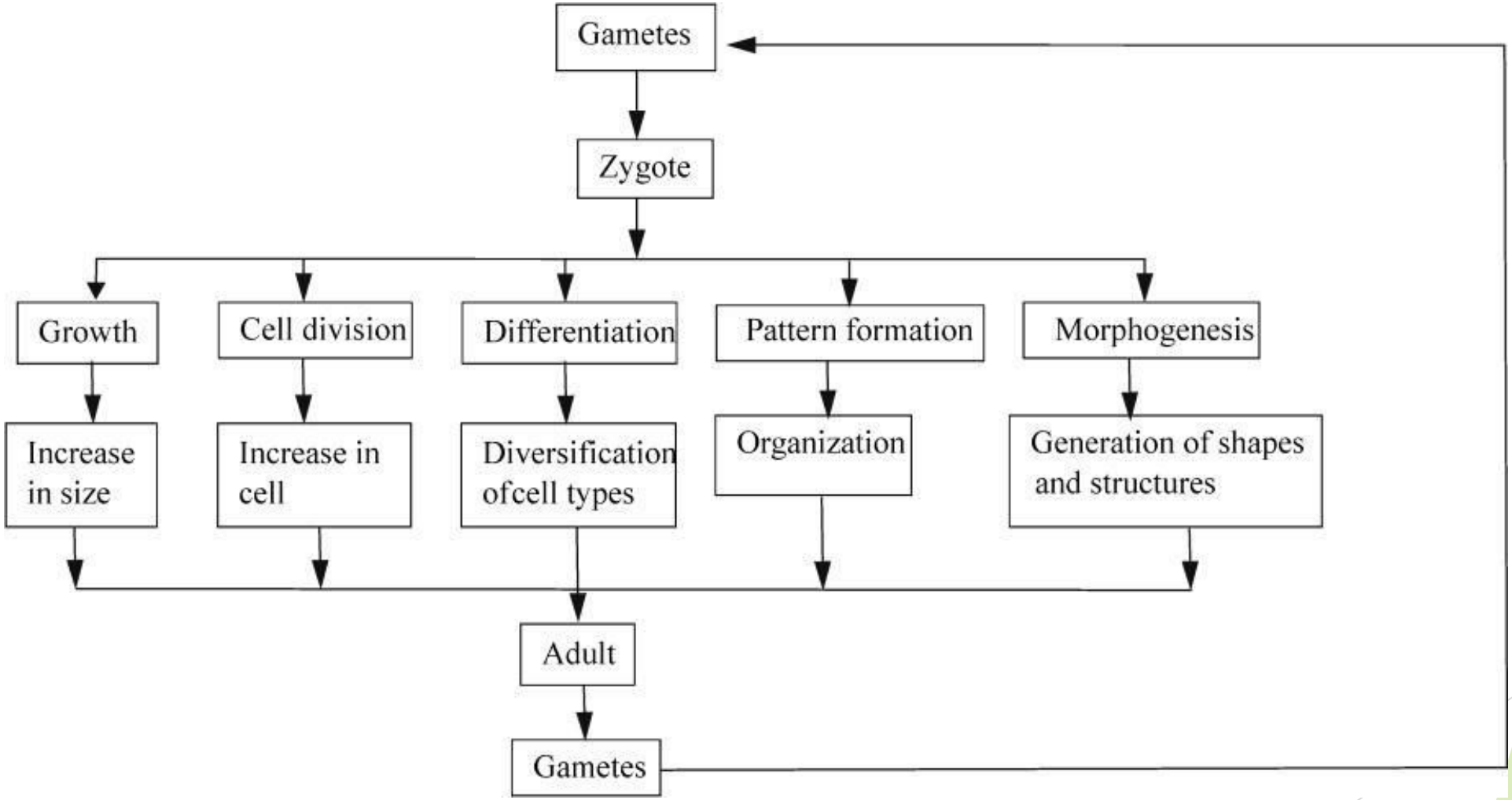
Each day we replace more than a gram of skin cells (the older cells being sloughed off as we move), and our bone marrow sustains the development of millions of new red blood cells every minute of our lives.

# Introduction to Developmental Biology

In addition, some animals can regenerate severed parts, and many species undergo metamorphosis (such as the transformation of a tadpole into a frog, or a caterpillar into a butterfly).

So **Developmental biology** is a discipline that studies embryonic and other developmental processes.

# Introduction to Developmental Biology



# Introduction to Developmental Biology

Development accomplishes two major objectives:

- It generates cellular diversity and order within each generation.
- It ensures the continuity of life from one generation to the next.

# Developmental hierarchy of animals

# Developmental hierarchy of animals

The **developmental hierarchy** is the hierarchical series of decisions involving specific pre-existing cell types.

These developmental decisions are usually irreversible, that progressively and irreversibly restrict cell fate.

# Developmental hierarchy of animals

Animal development is the division of embryo into the three germ layers.

1. **Ectoderm**
2. **Mesoderm**
3. **Endoderm**

# Developmental hierarchy of animals

## Potency and cell fate

Potency is the entire range of the cell types a particular cell can give rise to in all possible environments.

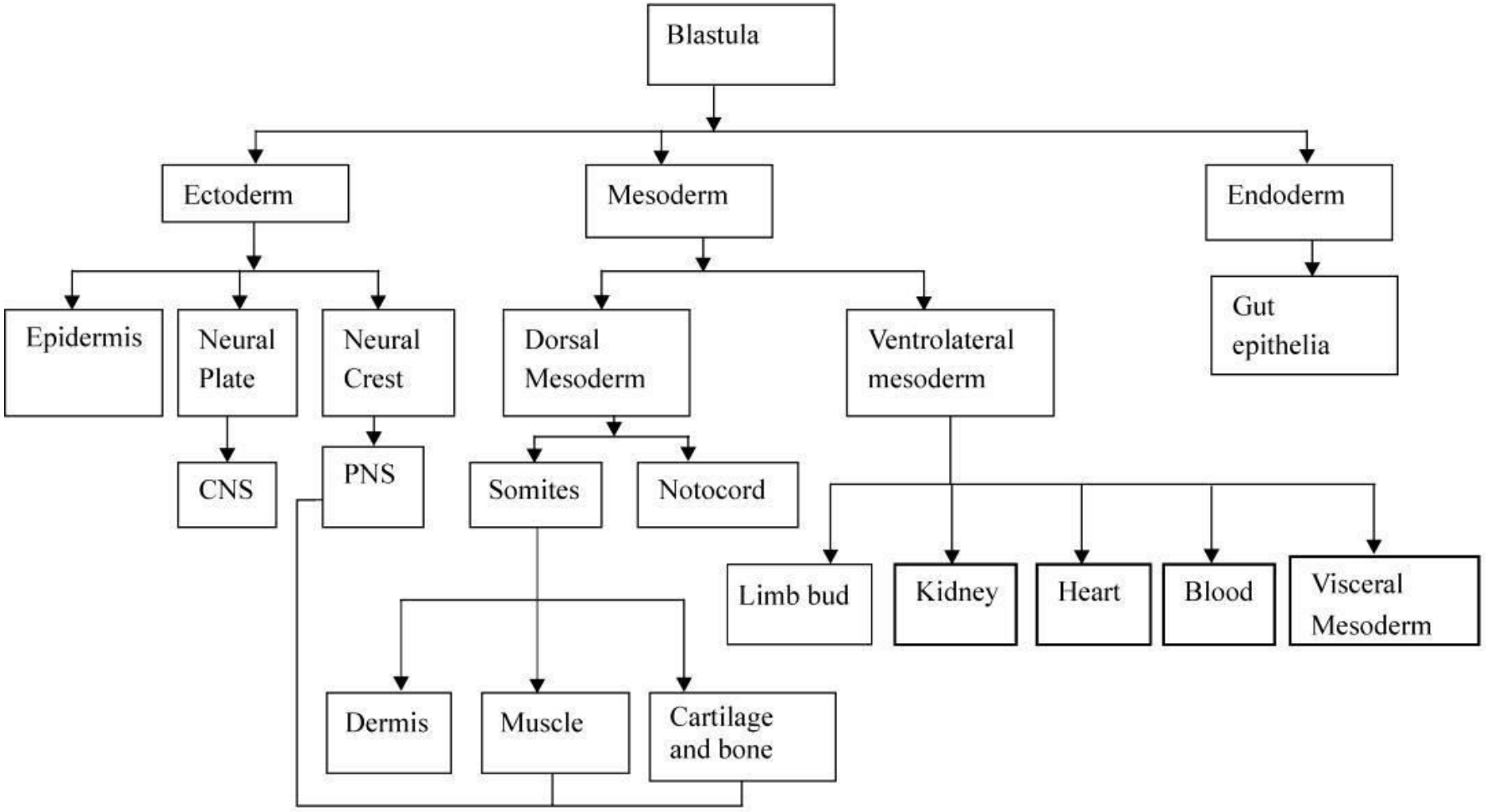
For example: A cell is pluripotent if it can give rise to all different blood cell lineages. There is a decrease in potency as cells become committed.

# Developmental hierarchy of animals

Fate of a cell is all the different cell types, its descendants can become during normal development.

- Cell potency ≥ fate
- Potency is intrinsic property.
- Cell fate is equal to potency plus environment.
- Cell fate becomes increasingly restricted until a cell is terminally differentiated.

# Developmental hierarchy of animals



# Anatomical Approaches to Developmental Biology

# Anatomical Approaches to Developmental Biology

The basis of all research in developmental biology is the changing anatomy of the organism. What parts of the embryo form the heart? How do the tissues that form the bird wing relate to the tissues that form the fish fin or the human hand?

# Anatomical Approaches to Developmental Biology

Following important parameters weave together to form the anatomical approaches to development.

- Comparative embryology
- Evolutionary embryology
- Teratology
- Mathematical modeling

# Anatomical Approaches to Developmental Biology

**Comparative embryology:**

The first strand is comparative embryology, the study of how anatomy changes during the development of different organisms. For instance, a comparative embryologist may study which tissues form the nervous system in the fly or in the frog.

# Anatomical Approaches to Developmental Biology

**Evolutionary embryology:**

The second strand, based on the first, is evolutionary embryology, the study of how changes in development may cause evolutionary changes and of how an organism's ancestry may constrain the types of changes that are possible.

# Anatomical Approaches to Developmental Biology

## Teratology:

The third anatomical approach to developmental biology is **teratology**, the study of birth defects. These anatomical abnormalities may be caused by mutant genes or by substances in the environment that interfere with development. The study of abnormalities is often used to discover how normal development occurs.

# Anatomical Approaches to Developmental Biology

**Mathematical modeling:**

The fourth anatomical approach is mathematical modeling, which seeks to describe developmental phenomena in terms of equations. Certain patterns of growth and differentiation can be explained by interactions whose results are mathematically predictable.

# Developmental Commitment

# Developmental Commitment

Before a cell overtly differentiates, it can be said to be committed, that is, the cell has been instructed on its fate. An intrinsic program has been activated within the cell that causes it to follow a particular pathway of development.

# Developmental Commitment

**Levels of developmental commitment**

- As cell fate becomes restricted following each decision in the developmental hierarchy, cells become committed to a certain fate.

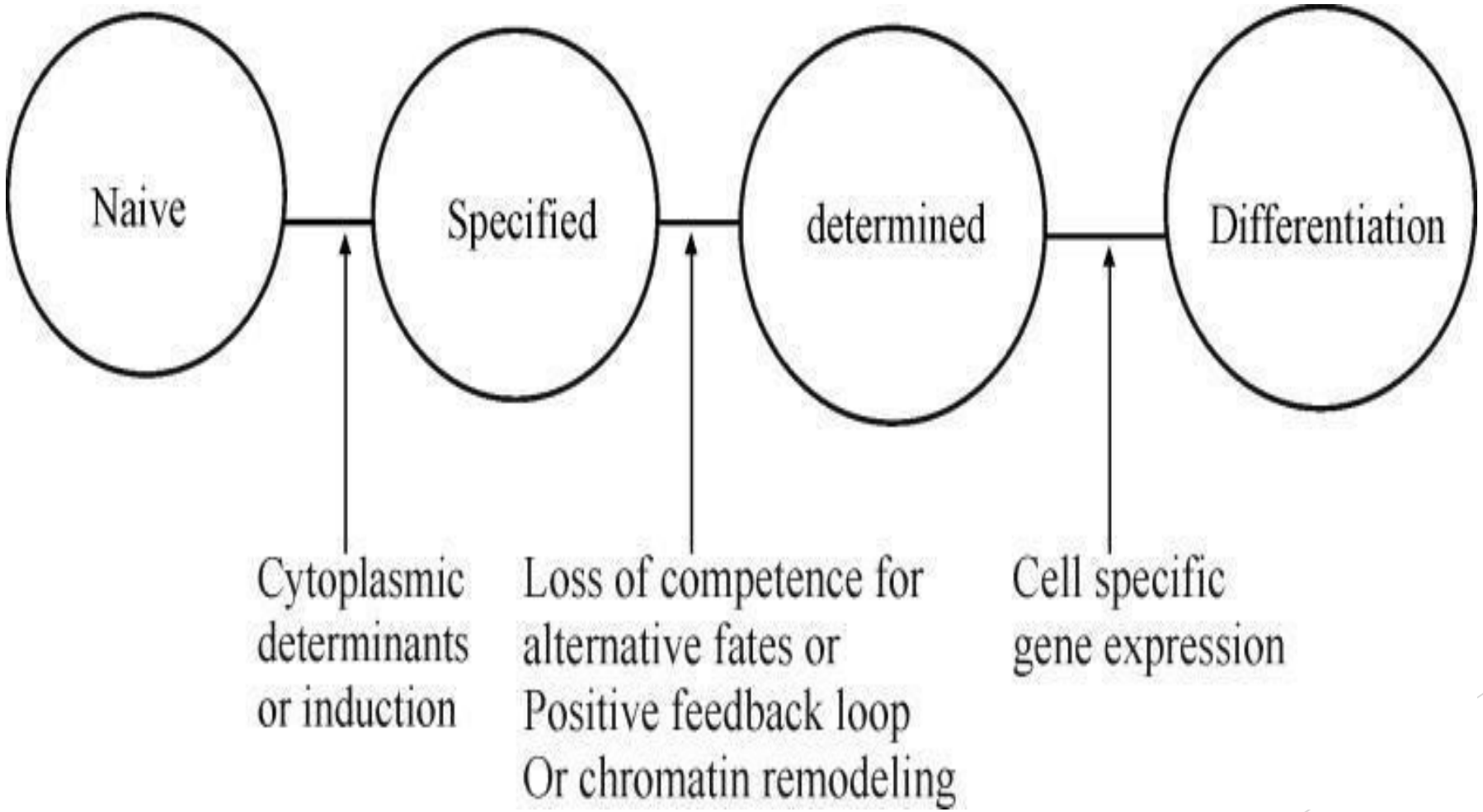
# Developmental Commitment

- An uncommitted cell can be described as naive, meaning that it has received no instructions directing it along a particular developmental pathway.
- The fate of a cell is said to be determined if it cannot be changed, regardless of cell's environment. Commitment at this stage becomes irreversible.

# Developmental Commitment

- Fate of a cell is said to be specified if the cell is directed to follow a certain developmental pathway and does so when placed in isolation, which should provide a neutral environment. Commitment at this stage is reversible as it may be specified if placed in different environment.

# Stages of Developmental Commitment



# Mechanisms of Developmental Commitment

# Mechanisms of Developmental Commitment

There appear to be two major strategies for establishing commitment and hence initiating the series of events that result in cell differentiation.

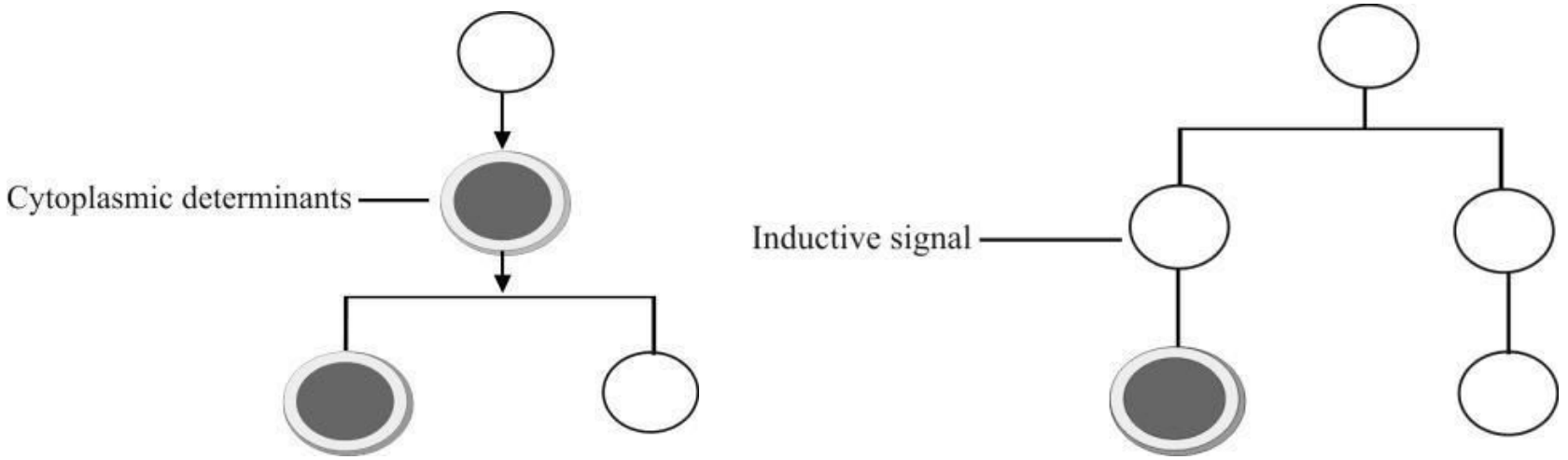
# Mechanisms of Developmental Commitment

- **The inheritance of cytoplasmic determinants:** Cytoplasmic determinants are the molecules in cytoplasm that can help to determine cell fate. The asymmetric distribution of cytoplasmic determinants indicates that the mechanism of differentiation is entirely intrinsic.

# Mechanisms of Developmental Commitment

For example, if a mother cell contains cytoplasmic determinants that is localized to one pole as the cell under goes division, that determinant will be inherited by only one of the daughters.

# Mechanisms of Developmental Commitment



# Mechanisms of Developmental Commitment

- **The perception of external inductive signals:** The process where one cell or group of cells changes developmental fate of another is termed induction. It is extrinsic process that depends on the position of a cell in the embryo.

# Mechanisms of Developmental Commitment

Two identical cells can follow alternative fates if one is exposed to an external signal (often secreted by a different cell).

# Modes of commitment

# Modes of commitment

Three basic modes of commitment are,

- **Autonomous specification**
- **Conditional specification**
- **Syncytial specification**

# Modes of commitment

**Autonomous specification:**

Specification by differential acquisition of certain cytoplasmic molecules present in the egg.

- Characteristic of most invertebrates.
- Cell type specification precedes any large-scale embryonic cell migration.

# Modes of commitment

- Autonomous specification gives rise to a pattern of development referred to as **mosaic development**, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts.

# Modes of commitment

- Invariant cleavages produce the same lineages in each embryo of the species. Blastomere fates are generally invariant.
- Produces "mosaic" ("determinative") development: cells cannot change fate if a blastomere is lost.

# Modes of commitment

**Conditional specification:**

- Characteristic of all vertebrates and few invertebrates.
- Specification by interactions between cells. Relative positions are important.

# Modes of commitment

- Variable cleavages produce no invariant fate assignments to cells.
- Massive cell rearrangements and migrations precede or accompany specification.
- Capacity for "regulative" development: allows cells to acquire different functions.

# Modes of commitment

**Syncytial specification:**

- Characteristic of most insect classes.
- Specification of body regions by interactions between cytoplasmic regions prior to cellularization of the blastoderm.

# Modes of commitment

- Variable cleavage produces no rigid cell fates for particular nuclei.
- After cellularization, conditional specification is most often seen.

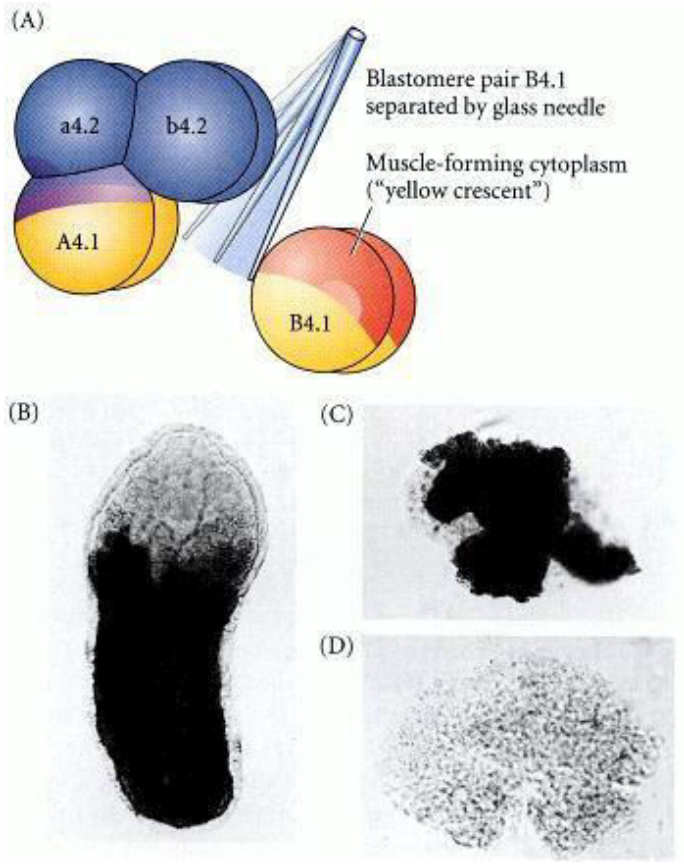
# Autonomous Specification

# Autonomous Specification

Autonomous specification was first demonstrated in 1887 by a French medical student, **Laurent Chabry**. Chabry desired to know the causes of birth defects, and he reasoned that such malformations might be caused by the lack of certain cells.

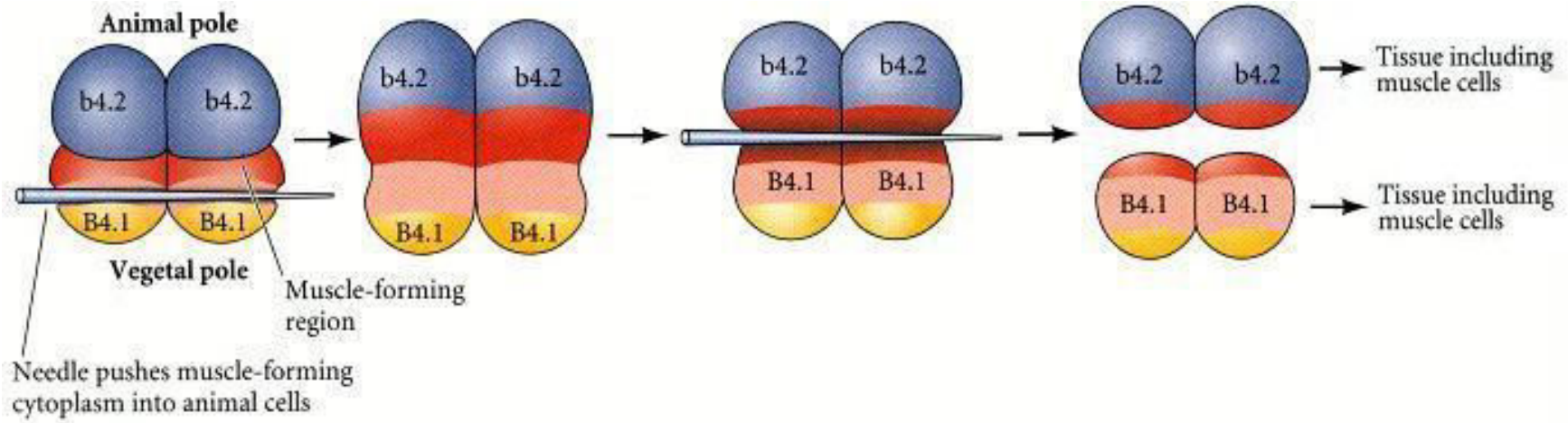
# Autonomous Specification

Chabry set out to produce specific malformations by isolating or lancing specific blastomeres of the cleaving tunicate embryo.



# Autonomous Specification

He discovered that each blastomere was responsible for producing a particular set of larval tissues.



# Autonomous Specification

In the absence of particular blastomeres, the larva lacked just those structures normally formed by those cells. Moreover, he observed that when particular cells were isolated from the rest of the embryo, they formed their characteristic structure apart from the context of the other cells. Thus, each of the tunicate cells appeared to be developing autonomously.

# Autonomous Specification

Recent studies have confirmed that when particular cells of the 8-cell tunicate embryo are removed, the embryo lacks those structures normally produced by the missing cells, and the isolated cells produce these structures away from the embryo.

# Autonomous Specification

**J. R. Whittaker** provided dramatic biochemical confirmation of the cytoplasmic segregation of the morphogenetic determinants responsible for this pattern. He stained blastomeres for the presence of the enzyme acetylcholinesterase.

# Conditional specification

# Conditional specification

This mode of commitment is sometimes called **conditional specification**, because the fate of a cell depends upon the conditions in which the cell finds itself.

This mode of commitment involves interactions with neighboring cells.

# Conditional specification

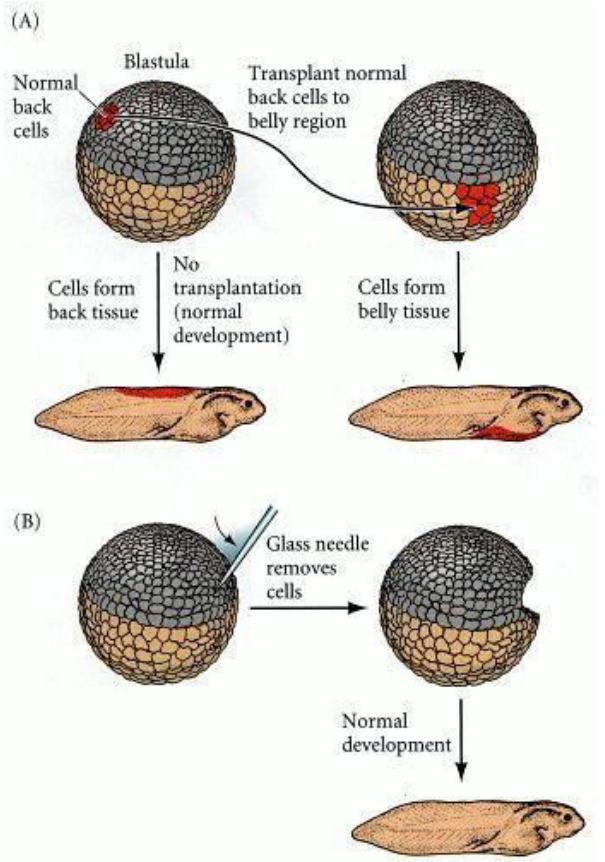
In this type of specification, each cell originally has the ability to become many different cell types. However, the interactions of the cell with other cells restricts the fate of one or both of the participants.

# Conditional specification

If a blastomere is removed from an early embryo that uses conditional specification, the remaining embryonic cells alter their fates so that the roles of the missing cells can be taken over.

# Conditional specification

This ability of the embryonic cells to change their fates to compensate for the missing parts is called **regulation.**



# Conditional specification

The isolated blastomere can also give rise to a wide variety of cell types (and sometimes generates cell types that the cell would normally not have made if it were part of the embryo).

Thus, conditional specification gives rise to a pattern of embryogenesis called **regulative development**.

# Conditional specification

Regulative development is seen in most vertebrate embryos, and it is obviously critical in the development of identical twins. In the formation of such twins, the cleavage-stage cells of a single embryo divide into two groups, and each group of cells produces a fully developed individual.

# Syncytial Specification

# Syncytial Specification

Many insects also use a third means, known as **Syncytial specification**, to commit cells to their fates. Here, interactions occur not between cells, but between parts of one cell.

# Syncytial Specification

In early embryos of these insects, cell division is not complete. Rather, the nuclei divide within the egg cytoplasm. This creates many nuclei in the large egg cell.

A cytoplasm that contains many nuclei is called a **syncytium**.

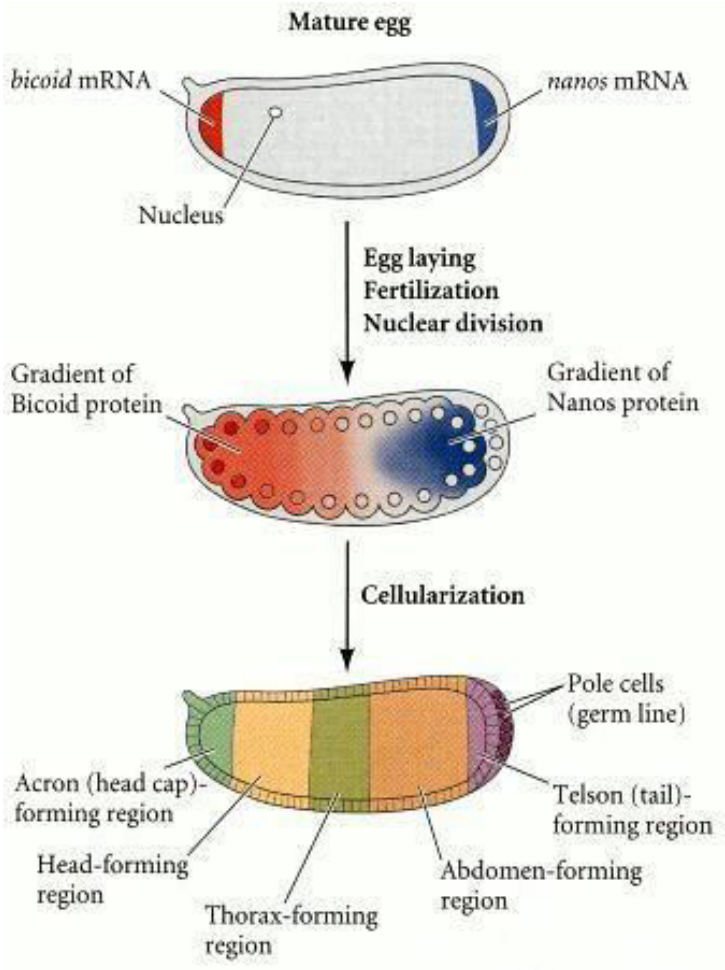
# Syncytial Specification

The egg cytoplasm, however, is not uniform.

Rather, the anterior of the egg cytoplasm is markedly different from the posterior. In *Drosophila,* for instance, the anterior most portion of the egg contains an mRNA that encodes a protein called **Bicoid**.

# Syncytial Specification

The posterior most portion of the egg contains an mRNA that encodes a protein called **Nanos.**



# Syncytial Specification

When the egg is laid and fertilized, these two mRNAs are translated into their respective proteins.

The **Bicoid** and **Nanos** proteins form a coordinate system based on their ratios, such that each region of the embryo will be distinguished by a different ratio of the two proteins.

# Syncytial Specification

- Those nuclei in regions containing high amounts of Bicoid and little Nanos will be instructed to activate those genes necessary for producing the head.
- Those nuclei in regions with slightly less Bicoid but with a small amount of Nanos will be instructed to activate those genes that generate the thorax.

# Syncytial Specification

- Those nuclei in regions that have little or no Bicoid and plenty of Nanos will be instructed to form the abdominal Structures.

Insects such as *Drosophila* use all three modes of specification to commit their cells to particular fates.

# Principles of Development

# Principles of Development

Following are the some important points that are considered as principles of development.

- Organisms must function as they form their organs. They have to use one set of structures while constructing others.

# Principles of Development

- Preformation is not in the anatomical structures, but in the instructions to form them. The inheritance of the fertilized egg includes the genetic potentials of the organism.
- The preformed nuclear instructions include the ability to respond to environmental stimuli in specific ways.

# Principles of Development

- The ectoderm gives rise to the epidermis, nervous system, and pigment cells.
- The mesoderm generates the kidneys, gonads, bones, heart, and blood cells.
- The endoderm forms the lining of the digestive tube and the respiratory system.

# Principles of Development

- **Karl von Baer's principles** state that the general features of a large group of animals appear earlier in the embryo than do the specialized features of a smaller group. The early embryo of a "higher" animal species is not like the adult of a "lower" animal.

# Principles of Development

- Homologous structures in different species are those organs whose similarity is due to their sharing a common ancestral structure. Analogous structures are those organs whose similarity comes from their serving a similar function (but which are not derived from a common ancestral structure).

# Principles of Development

- "Community of embryonic structure reveals community of descent" (Charles Darwin).
- Congenital anomalies can be caused by genetic factors (mutations, aneuploidies, translocations) or by environmental agents (certain chemicals, certain viruses, radiation).

# Principles of Development

- Syndromes consists of sets of developmental abnormalities that "run together."
- Organs that are linked in developmental syndromes share either a common origin or a common mechanism of formation.
- Allometric growth can create dramatic changes in the structure of organisms.

# The Stages of Animal Development

# The Stages of Animal Development

**Cleavage:**

Cleavage is a series of extremely rapid mitotic divisions wherein the enormous volume of zygote cytoplasm is divided into numerous smaller cells. These cells are called **blastomeres**, and by the end of cleavage, they generally form a sphere known as a **blastula**.

# The Stages of Animal Development

**Gastrulation:**

A series of extensive cell rearrangements is called **gastrulation,** and the embryo is said to be in the **gastrula** stage. As a result of gastrulation, the embryo contains three **germ layers**: the ectoderm, the endoderm, and the mesoderm.

# The Stages of Animal Development

**Organogenesis**:

Once the three germ layers are established, the cells interact with one another and rearrange themselves to produce tissues and organs. This process is called **organogenesis**. Many organs contain cells from more than one germ layer.

# The Stages of Animal Development

For example, the outer layer of skin comes from the ectoderm, while the inner layer (the dermis) comes from the mesoderm. Also during organogenesis, certain cells undergo long migrations from their place of origin to their final location.

# The Stages of Animal Development

**Germ cells and somatic cells:**

In many species a specialized portion of egg cytoplasm gives rise to cells that are the precursors of the **gametes** (the sperm and egg). The gametes and their precursor cells are collectively called **germ cells**, and they are set aside for reproductive function.

# The Stages of Animal Development

All the other cells of the body are called **Somatic cells**. This separation of somatic cells (which give rise to the individual body) and germ cells (which contribute to the formation of a new generation) is often one of the first differentiations to occur during animal development.

# The Stages of Animal Development

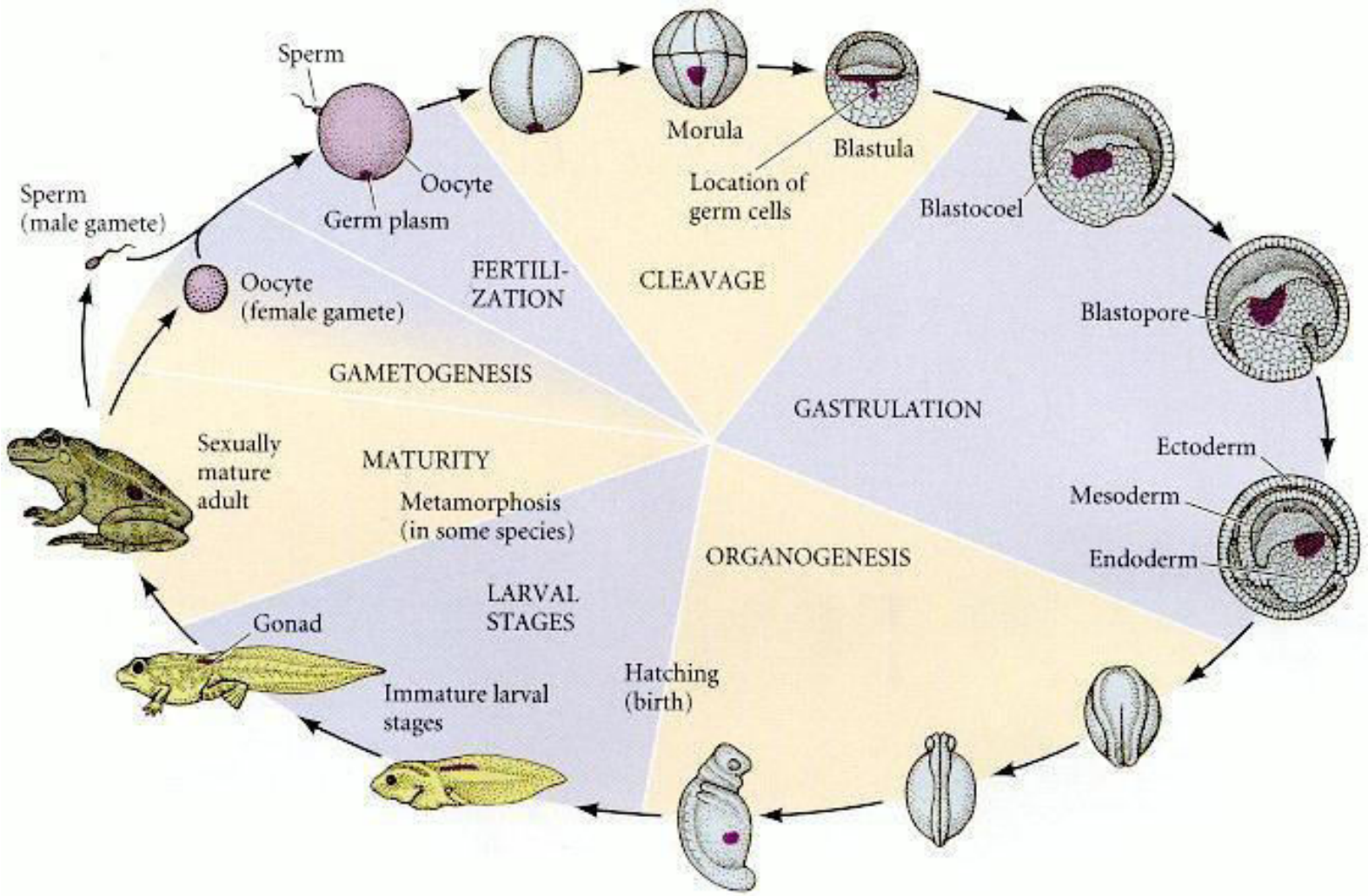
**Gametogenesis:**

The germ cells eventually migrate to the gonads, where they differentiate into gametes. The development of gametes, called **gametogenesis**, is usually not completed until the organism has become physically mature.

# The Stages of Animal Development

In many species, the organism that hatches from the egg or is born into the world is not sexually mature. That young organism is a **larva** that may look significantly different from the adult. Larvae often constitute the stage of life that is used for feeding or dispersal.

# The Stages of Animal Development



# Amphibian Development

# Amphibian Development

In amphibians, metamorphosis is initiated by hormones from the tadpole's thyroid gland, and these changes prepare an aquatic organism for a terrestrial existence.

# Amphibian Development

In anurans (frogs and toads), the metamorphic changes are most striking, and almost every organ is subject to modification.

- For locomotion, the hind limbs and forelimbs differentiate as the paddle tail withdraws.
- The gills regress, and the lungs enlarge.

# Amphibian Development

- The horny teeth the tadpole uses to tear up pond plants disappear as the mouth and jaw take a new shape.
- Meanwhile the large intestine characteristic of herbivores shortens to suit the more carnivorous diet of the adult frog.

# Amphibian Development

- As metamorphosis ends, the development of the first germ cells begins. In *Rana pipiens*, egg development lasts 3 years.
- The speed of metamorphosis is directly linked with environmental pressures.

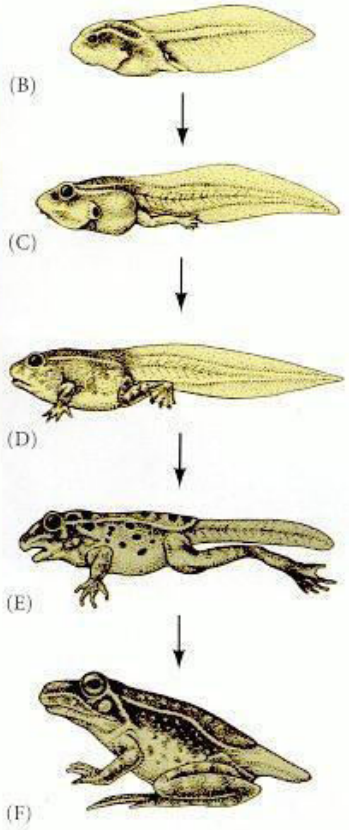
# Amphibian Development

- Since the bottom half of the egg usually contains the yolk, it divides more slowly (because the large yolk deposits interfere with cleavage). This portion is the **vegetal** hemisphere of the egg.

# Amphibian Development

- Conversely, the upper half of the egg usually has less yolk and divides faster. This upper portion is called the **animal** hemisphere of the egg.


(A)



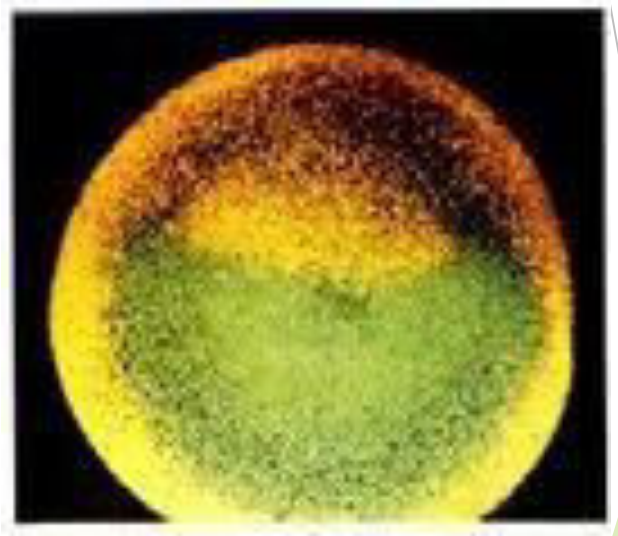
# Amphibian Development

Since the bottom half of the egg usually contains the yolk, it divides more slowly (because the large yolk deposits interfere with cleavage). This portion is the **vegetal** hemisphere of the egg. Conversely, the upper half of the egg usually has less yolk and divides faster. This upper portion is called the **animal** hemisphere of the egg.

# The Frog Life Cycle

# The Frog Life Cycle

If the frog is mature, the pituitary gland secretes hormones that stimulate the ovary to make estrogen. **Estrogen** is a hormone that can instruct the liver to make and secrete the yolk proteins, which are then transported through the blood into the enlarging eggs in the ovary. The yolk is transported into the bottom portion of the egg.



# The Frog Life Cycle

Another ovarian hormone, **progesterone**, signals the egg to resume its meiotic division. This is necessary because the egg had been "frozen" in the metaphase of its first meiosis. When it has completed this first meiotic division, the egg is released from the ovary and can be fertilized.

# The Frog Life Cycle

The male leopard frogs make their sperm in the summer, and by the time they begin hibernation in autumn, they have all the sperm that are to be available for the following spring's breeding season.

# The Frog Life Cycle

In most species of frogs, fertilization is external. The male frog grabs the female's back and fertilizes the eggs as the female frog releases them.



# The Frog Life Cycle

***Rana pipiens*** usually lays around 2500 eggs, while the bullfrog, ***Rana catesbiana***, can lay as many as 20,000. Some species lay their eggs in pond vegetation, and the jelly adheres to the plants and anchors the eggs.



# The Frog Life Cycle

Fertilization accomplishes several things.

- First, it allows the egg to complete its second meiotic division, which provides the egg with a haploid **pronucleus**. The egg pronucleus and the sperm pronucleus will meet in the egg cytoplasm to form the diploid zygotic nucleus.

# The Frog Life Cycle

- Second, fertilization causes the cytoplasm of the egg to move such that different parts of the cytoplasm find themselves in new locations.

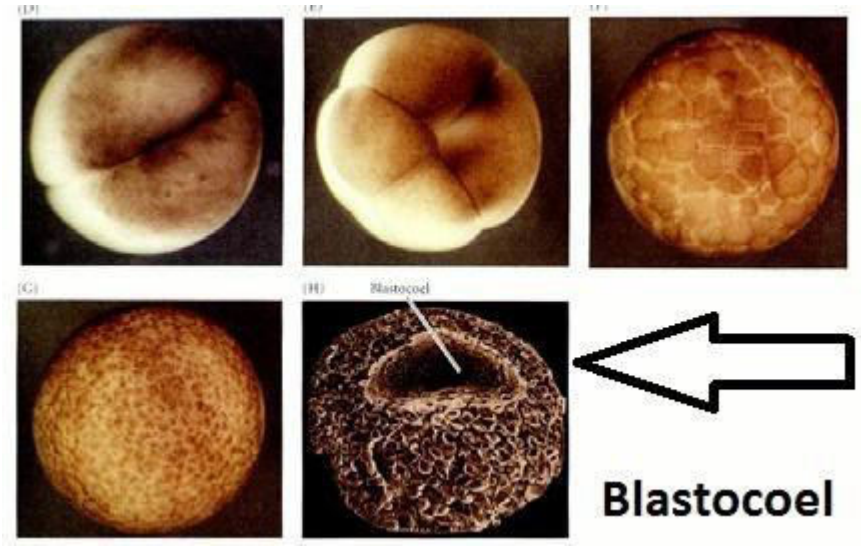


# The Frog Life Cycle

- Third, fertilization activates those molecules necessary to begin cell cleavage and development. The sperm and egg die quickly unless fertilization occurs.

# The Frog Life Cycle

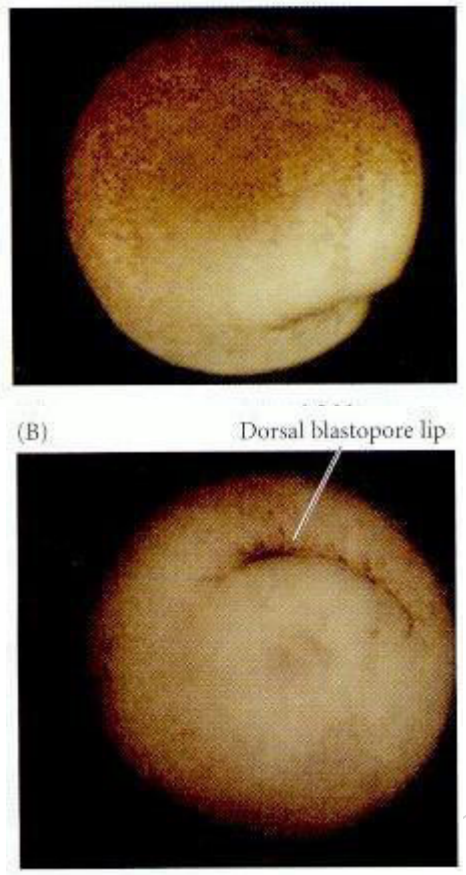
During cleavage, the volume of the frog egg stays the same, but it is divided into tens of thousands of cells. A fluid-filled cavity, the **blastocoel**, forms in the animal hemisphere. This cavity will be important for allowing cell movements to occur during gastrulation.



# The Frog Life Cycle II

# The Frog Life Cycle II

Gastrulation in the frog begins at a point on the embryo surface roughly 180 degrees opposite the point of sperm entry with the formation of a dimple, called the **blastopore.** Cells migrate through the blastopore and toward the animal pole. These cells become the dorsal mesoderm.

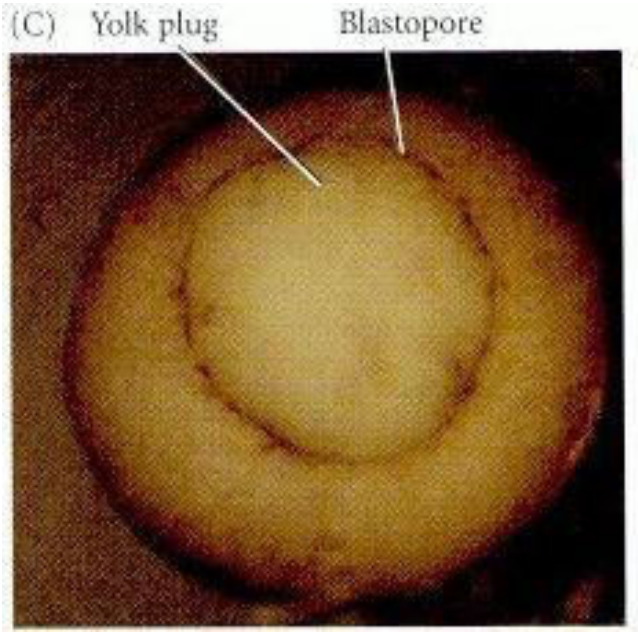


# The Frog Life Cycle II

The blastopore expands into a circle and cells migrating through this circle become the **lateral and ventral mesoderm.**

The cells remaining on the outside become the **ectoderm.**

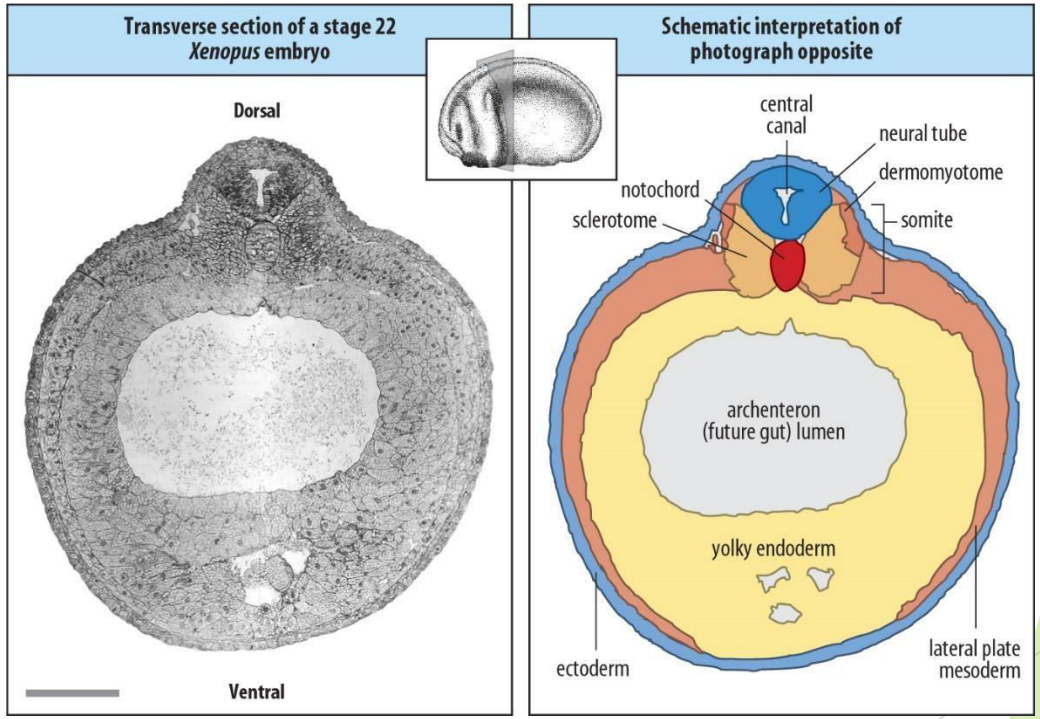
The large yolky cells that remain at the vegetal hemisphere (until they are encircled by the ectoderm) become the **endoderm.**



# The Frog Life Cycle II

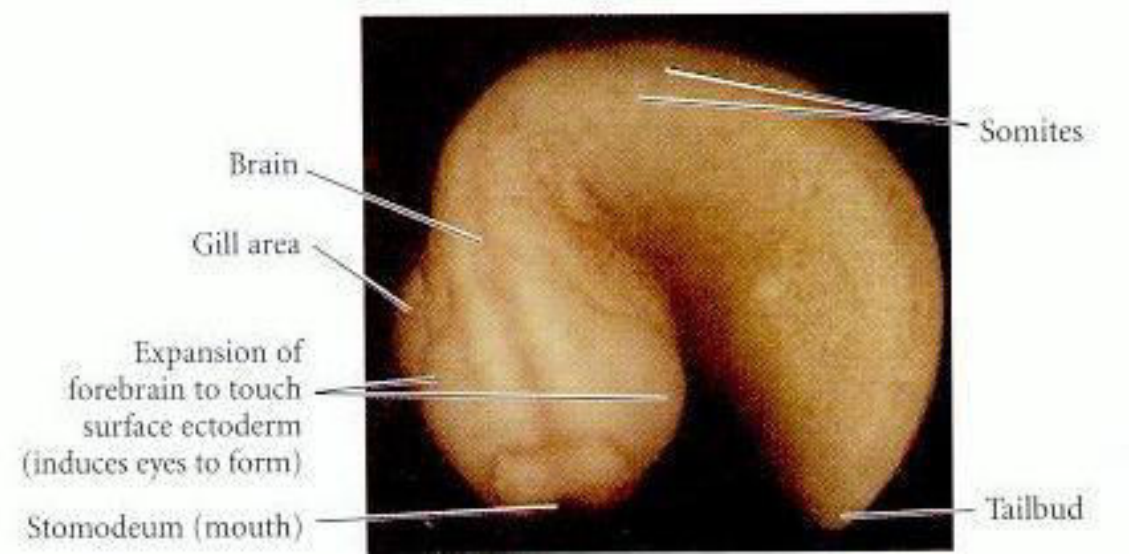
## Notochord:

A rod of mesodermal cells in the most dorsal portion of the embryo. At this stage, the embryo is called a **Neurula**. The neural precursor cells elongate, stretch, and fold into the embryo forming the **neural tube**.



# The Frog Life Cycle II

Once the neural tube has formed, it induces changes in its neighbors, and organogenesis continues. The mesodermal tissue adjacent to the notochord becomes segmented into **somites**, the precursors of the frog's back muscles, spinal cord, and dermis (the inner portion of the skin).



# The Frog Life Cycle II

**Tadpole:**

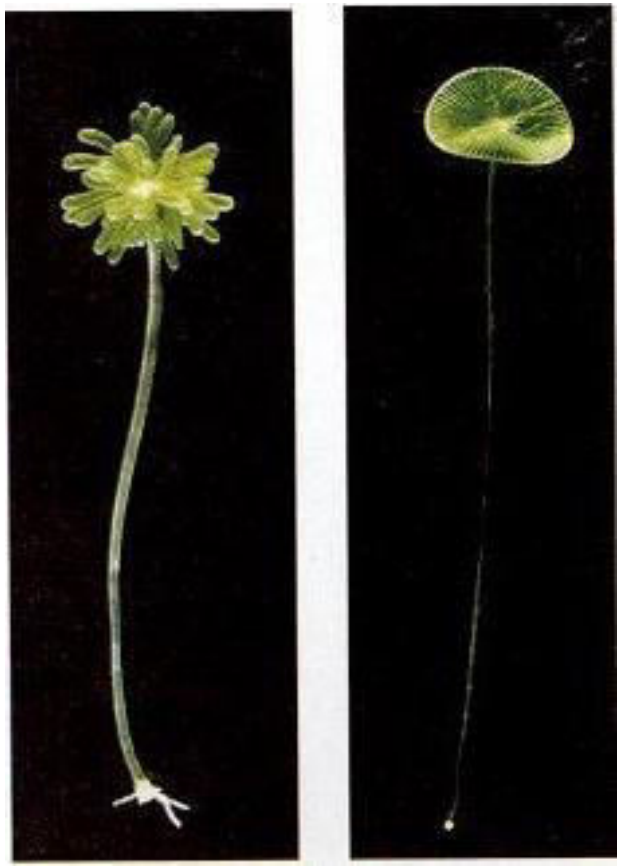
Somites appear as blocks of mesodermal tissue. The embryo develops a mouth and an anus, and it elongates into the typical tadpole structure.



# The role of nucleus in Morphogenesis

# The role of nucleus in Morphogenesis

Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in *Acetabularia.* This enormous single cell (2-4 cm long) consists of three parts:



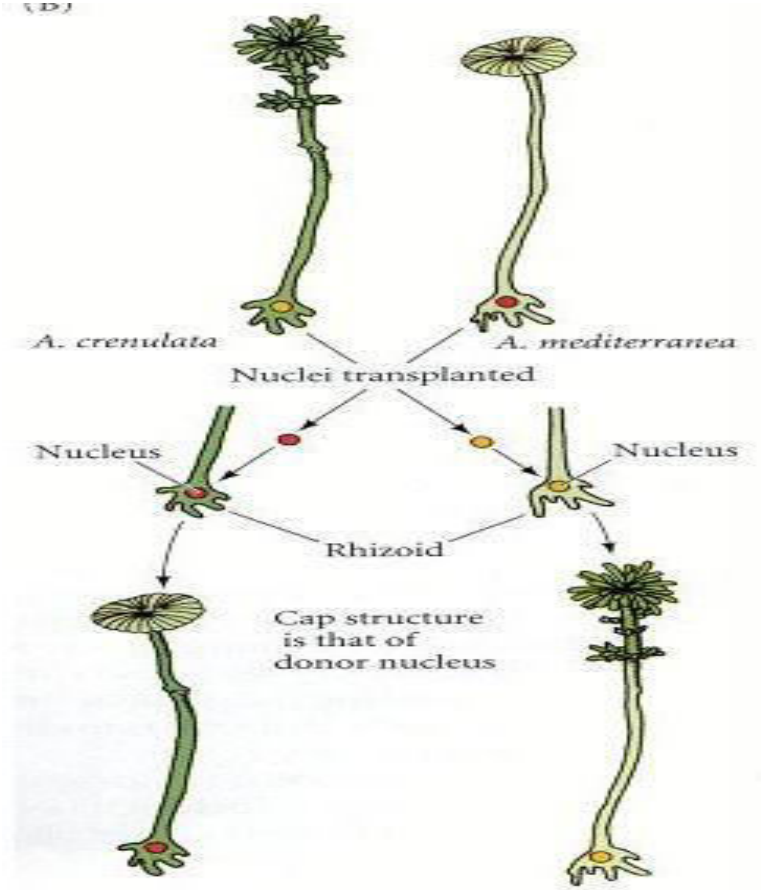
- a cap,
- a stalk,
- a rhizoid.

# The role of nucleus in Morphogenesis

The single nucleus of the cell resides within the rhizoid. The size of *Acetabularia* and the location of its nucleus allow investigators to remove the nucleus from one cell and replace it with a nucleus from another cell.

# The role of nucleus in Morphogenesis

**J.Hammerling** exchanged nuclei between two morphologically distinct species, *A. mediterranea* and *A. crenulata.* Figure shows that these two species have very different cap structures.

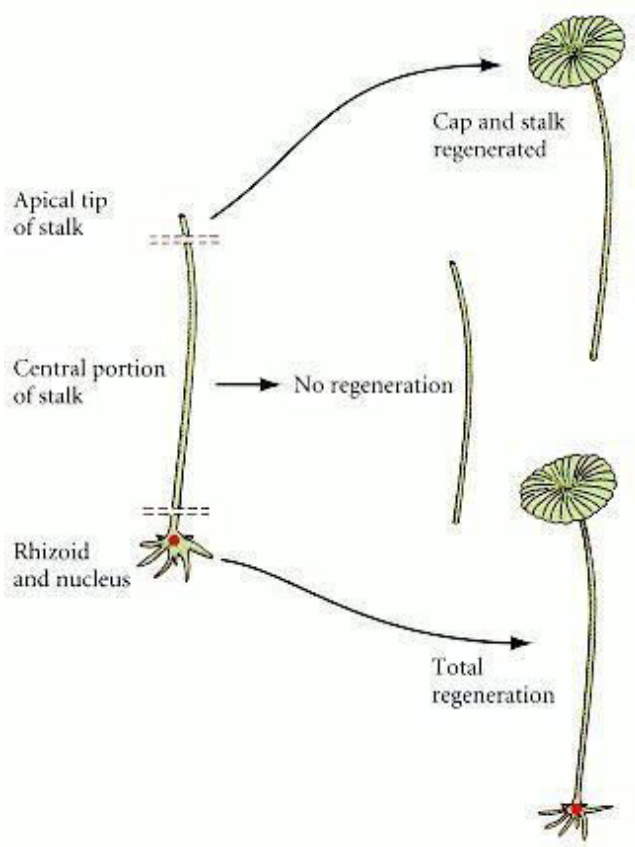


# The role of nucleus in Morphogenesis

**J. Hammerling** found that when he transferred the nucleus from one species into the stalk of another species, the newly formed cap eventually assumed the form associated with the *donor* nucleus. Thus, the nucleus was seen to control *Acetabularia* development.

# The role of nucleus in Morphogenesis

The formation of a cap is a complex morphogenic event involving the synthesis of numerous proteins, which must be accumulated in a certain portion of the cell and then assembled into complex, species-specific structures.



# The role of nucleus in Morphogenesis

These studies suggest that

- Nucleus contains information specifying the type of cap produced (i.e., it contains the genetic information that specifies the proteins required for the production of a certain type of cap).
- Material containing this information enters the cytoplasm long before cap production occurs.

# The role of nucleus in Morphogenesis

One hypothesis proposed on the base of Hammerling study is that,

“The nucleus synthesizes a stable mRNA that lies dormant in the cytoplasm until the time of cap formation”.

# The role of nucleus in Morphogenesis

Hence, the expression of the cap is controlled not only by nuclear transcription, but also by the translation of the cytoplasmic RNA. In this unicellular organism, "development" is controlled at both the transcriptional and translational levels.

# Unicellular protists and the origins of sexual reproduction

# Unicellular protists and the origins of sexual reproduction

Sex and reproduction are two distinct and separable processes. **Reproduction** involves the creation of new individuals.

**Sex** involves the combining of genes from two different individuals into new arrangements.

# Unicellular protists and the origins of sexual reproduction

Reproduction in the absence of sex is characteristic of organisms that reproduce by fission (i.e., splitting into two); there is no sorting of genes when an amoeba divides or when a hydra buds off cells to form a new colony.

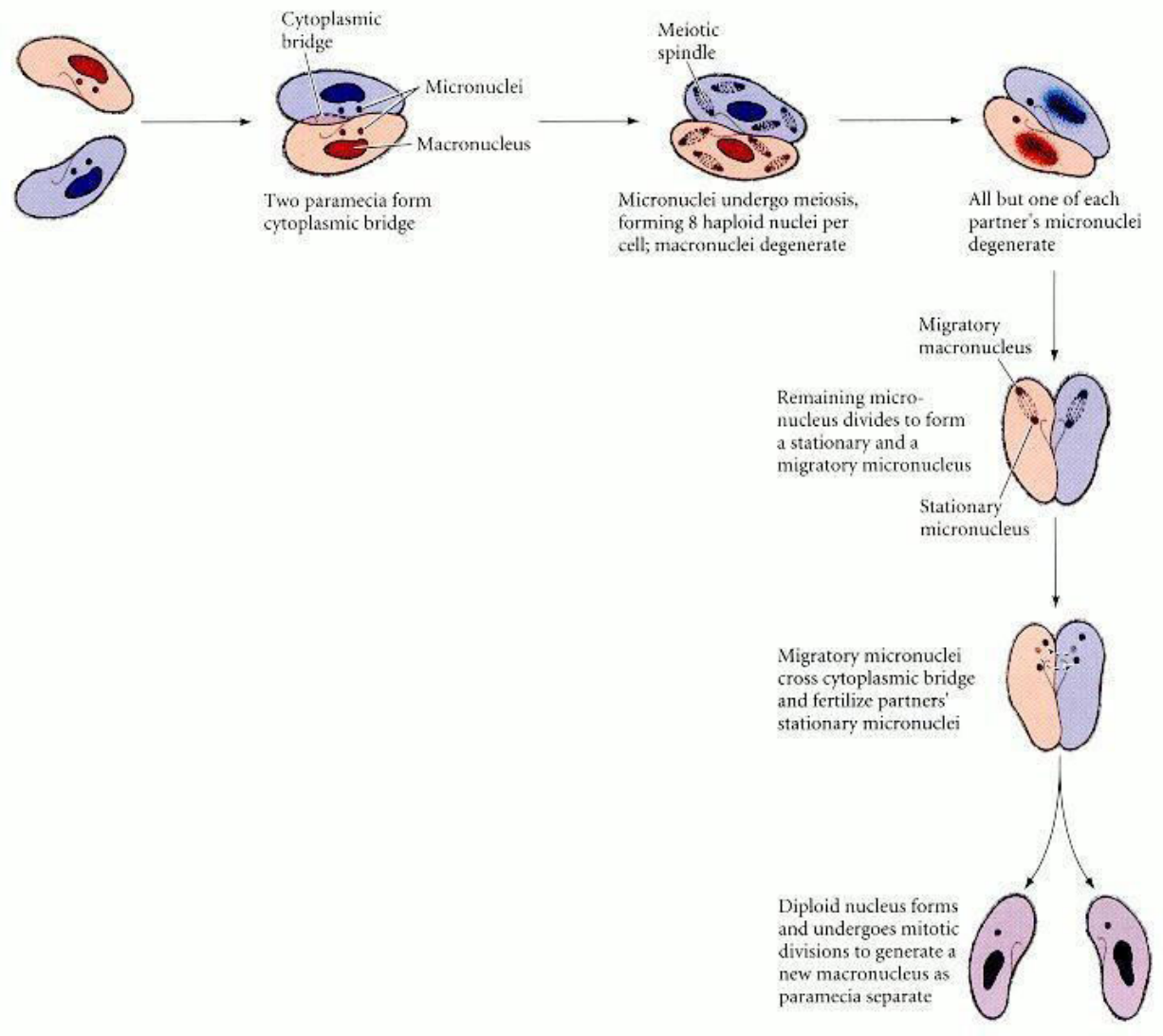
# Unicellular protists and the origins of sexual reproduction

Sex without reproduction is also common among unicellular organisms.

- Bacteria are able to transmit genes from one individual to another by means of sex pili. This transmission is separate from reproduction.
- Protists are also able to reassort genes without reproduction. Paramecia, for instance, reproduce by fission, but sex is accomplished by **conjugation**.

# Unicellular protists and the origins of sexual reproduction

When two paramecia join together, they link their oral apparatuses and form a cytoplasmic connection through which they can exchange genetic material



# Unicellular protists and the origins of sexual reproduction

Each macronucleus (which controls the metabolism of the organism) degenerates, while each micronucleus undergoes meiosis to produce eight haploid micronuclei, of which all but one degenerate. The remaining micronucleus divides once more to form a stationary micronucleus and a migratory micronucleus.

# Unicellular protists and the origins of sexual reproduction

Each migratory micronucleus crosses the cytoplasmic bridge and fuses with ("fertilizes") the stationary micronucleus, thereby creating a new diploid nucleus in each cell. This diploid nucleus then divides mitotically to give rise to a new micronucleus and a new macronucleus as the two partners disengage. Therefore, no reproduction has occurred, only sex.

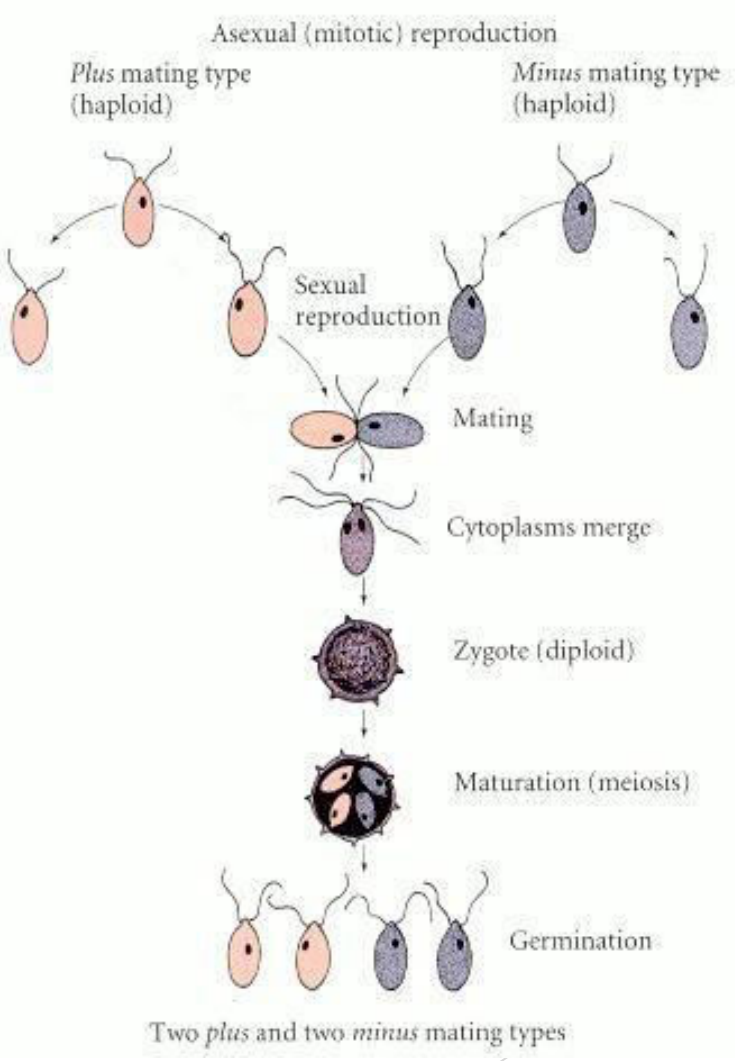
# Sexual Reproduction in Unicellular Eukaryotes

# Sexual Reproduction in Unicellular Eukaryotes

The union of these two distinct processes, sex and reproduction, into **sexual reproduction** is seen in unicellular eukaryotes. Let us take an example of the life cycle of *Chlamydomonas*. Which is usually haploid, having just one copy of each chromosome.

# Sexual Reproduction in Unicellular Eukaryotes

The individuals of each species, however, are divided into two mating types: plus and minus. When a plus and a minus meet, they join their cytoplasm, and their nuclei fuse to form a diploid zygote.



# Sexual Reproduction in Unicellular Eukaryotes

The flagella of two individuals twist around each other, enabling specific regions of the cell membranes to come together. These specialized regions contain mating type-specific components that enable the cytoplasms to fuse.

# Sexual Reproduction in Unicellular Eukaryotes

Following flagellar agglutination, the plus individuals initiate fusion by extending a fertilization tube.

This tube contacts and fuses with a specific site on the minus individual.

# Sexual Reproduction in Unicellular Eukaryotes

In evolving sexual reproduction, two important advances had to be achieved.

- The first was the mechanism of meiosis whereby the diploid complement of chromosomes is reduced to the haploid state.
- The second was a mechanism whereby the two different mating types could recognize each other.

# Developmental Patterns among the Metazoa

# Developmental Patterns among the Metazoa

Metazoans belong to one of three major branches:

- Diploblasts,
- Protostomes,
- Deuterostomes.

# Developmental Patterns among the Metazoa

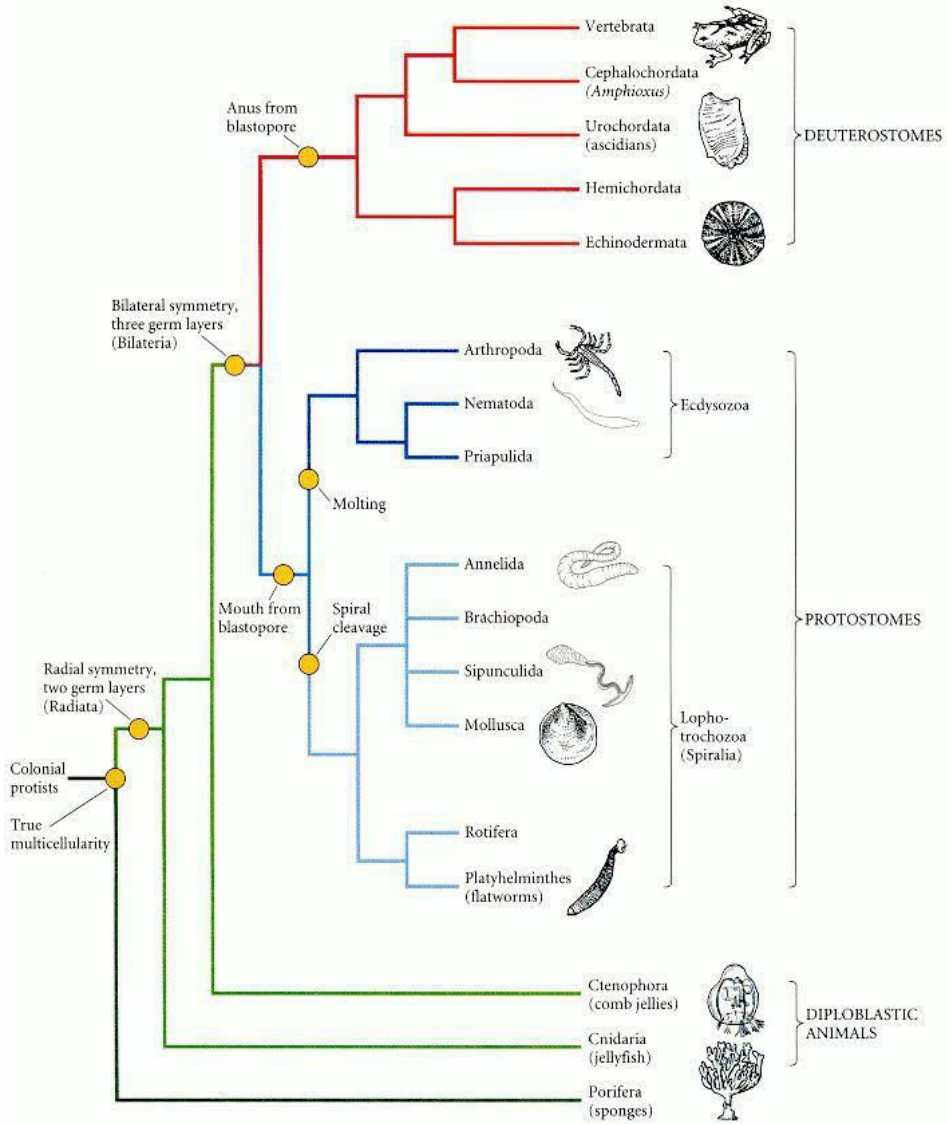
Sponges develop in a manner so different from that of any other animal group that some taxonomists do not consider them metazoans at all, and call them "**Parazoans.**"

# Developmental Patterns among the Metazoa

## Diploblasts:

Diploblastic animals are those who have ectoderm and endoderm, but no true mesoderm.

- These include the cnidarians (jellyfish and hydras) and the ctenophores (comb jellies).



# Developmental Patterns among the Metazoa

**Protostomes:**

(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location.

# Developmental Patterns among the Metazoa

There are two major branches of the protostomes.

- The **Ecdysozoa** includes those animals that molt. Its major constituent is Arthropoda, a phylum containing insects, arachnids, mites, crustaceans, and millipedes.
- The second major group of protostomes are the **Lophotrochozoa**. They are characterized by a common type of cleavage (spiral), a common larval form, and a distinctive feeding apparatus. These phyla include annelids, molluscs, and flatworms.

# Developmental Patterns among the Metazoa

**Deuterostomes:**

In Deuterostomes ("mouth second"), the mouth opening is formed after the anal opening.

Phyla in the Deuterostomes lineage include the chordates and echinoderms.

# Cadherins and cell adhesion

# Cadherins and cell adhesion

Recent evidence shows that boundaries between tissues can indeed be created both by,

- Different cell types having different types of cell adhesion molecules.
- Different cell types having different amounts of cell adhesion molecules.

# Cadherins and cell adhesion

There are several classes of molecules that can mediate cell adhesion. The major cell adhesion molecules appear to be the **Cadherins**.

As their name suggests, they are calcium-dependent adhesion molecules.

# Cadherins and cell adhesion

Cadherins are critical for establishing and maintaining intercellular connections, and they appear to be crucial to the spatial segregation of cell types and to the organization of animal form.

# Cadherins and cell adhesion

Cadherins interact with other Cadherins on adjacent cells, and they are anchored into the cell by a complex of proteins called **Catenins**. The cadherin-catenin complex forms the classic adherens junctions that connect epithelial cells together.

# Cadherins and cell adhesion

In vertebrate embryos, several major cadherin classes have been identified:

- **E-cadherin**
- **P-cadherin**
- **N-cadherin**
- **Protocadherins**

Effort By: AARISH ALI

# Cadherins and cell adhesion

**E-cadherin:**

Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the 1-cell stage. Later, this molecule is restricted to epithelial tissues of embryos and adults.

# Cadherins and cell adhesion

**P-cadherin:**

Placental cadherin appears to be expressed primarily on the trophoblast cells (those placental cells of the mammalian embryo that contact the uterine wall) and on the uterine wall epithelium.

# Cadherins and cell adhesion

**N-cadherin:**

Neural cadherin is first seen on mesodermal cells in the gastrulating embryo as they lose their E-cadherin expression. It is also highly expressed on the cells of the developing central nervous system.

# Cadherins and cell adhesion

**Protocadherins:**

Calcium-dependent adhesion proteins that differ from the classic Cadherins in that they lack connections to the cytoskeleton through catenins. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during *Xenopus* gastrulation.

# Genes and Development

# Genes and Development

Following important points depict the relationship of genes and development.

- Development connects genotype and phenotype.
- Nuclear genes are not lost or mutated during development. The genome of every cell is equivalent.

# Genes and Development

- The exceptions to the rule of genomic equivalence are the lymphocytes. During differentiation, these cells rearrange their DNA to create new immunoglobulin and antigen receptor genes.

# Genes and Development

- The ability of nuclei from differentiated cells to direct the development of complete adult organisms has recently confirmed the principle of genomic equivalence.
- Only a small percentage of the genome is expressed in any particular cell.

# Genes and Development

- The cloning of human beings, as well as regenerating damaged organs or enhancing physical abilities, may soon be possible through cloning technology and the use of embryonic stem cells.
- Northern blots, in situ hybridization, and the polymerase chain reaction can show which cells are transcribing particular genes.

# Genes and Development

- The functions of a gene often can be ascertained by antisense mRNA, transgenic expression, or (in the case of mammals) gene knockouts.
- Knowledge of gene activity in humans can be obtained by candidate gene mapping or positional cloning.

# Differential gene expression

# Differential gene expression

Different cell types use different subsets of these genes. Red blood cells make globins, lens cells make crystallins, melanocytes make melanin, and endocrine glands make their specific hormones.

# Differential gene expression

**Developmental genetics** is the discipline that examines how the genotype is transformed into the phenotype, and the major paradigm of developmental genetics is differential gene expression from the same nuclear repertoire.

# Differential gene expression

The regulation of gene expression can be accomplished at several levels:

- **Differential gene transcription:**

It regulates that which of the nuclear genes are transcribed into RNA.

# Differential gene expression

- **Selective nuclear RNA processing:**

It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.

# Differential gene expression

- **Selective messenger RNA translation:**

It regulates that which of the mRNAs in the cytoplasm become translated into proteins.

- **Differential protein modification:**

It regulates that which proteins are allowed to remain or function in the cell.

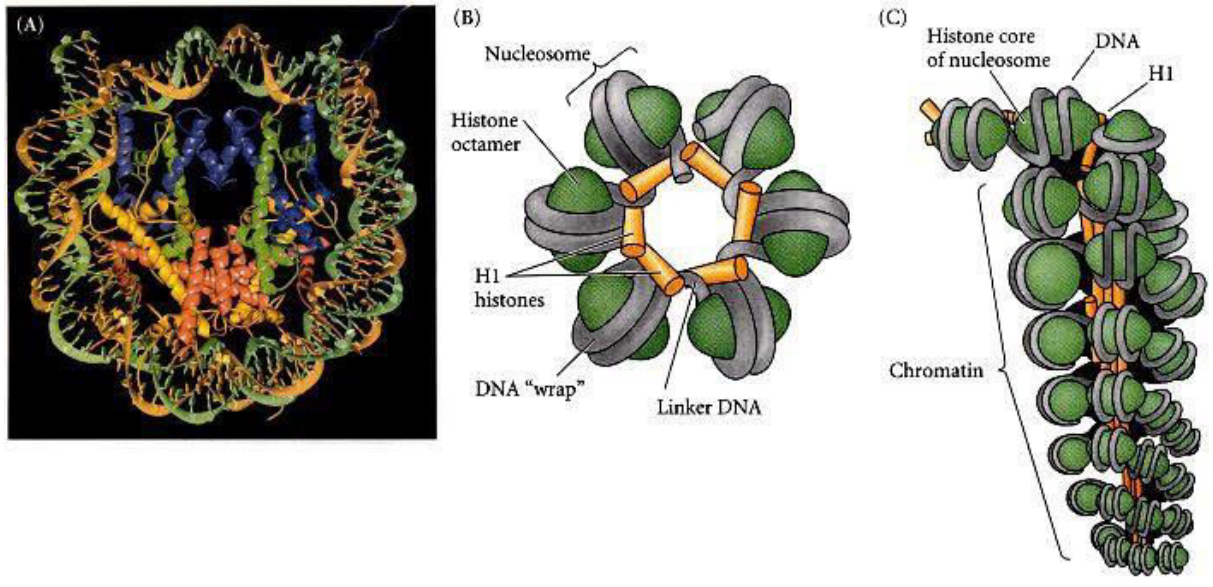
# Anatomy of the gene

# Anatomy of the gene

Eukaryotic genes are contained within a complex of DNA and protein called **chromatin**. The protein component constitutes about half the weight of chromatin and is composed largely of **nucleosomes**.

# Anatomy of the gene

**Nucleosome is** composed of an octamer of **histone** proteins (two molecules each of histones H2A-H2B and histones H3-H4) wrapped with two loops containing approximately 140 base pairs of DNA.



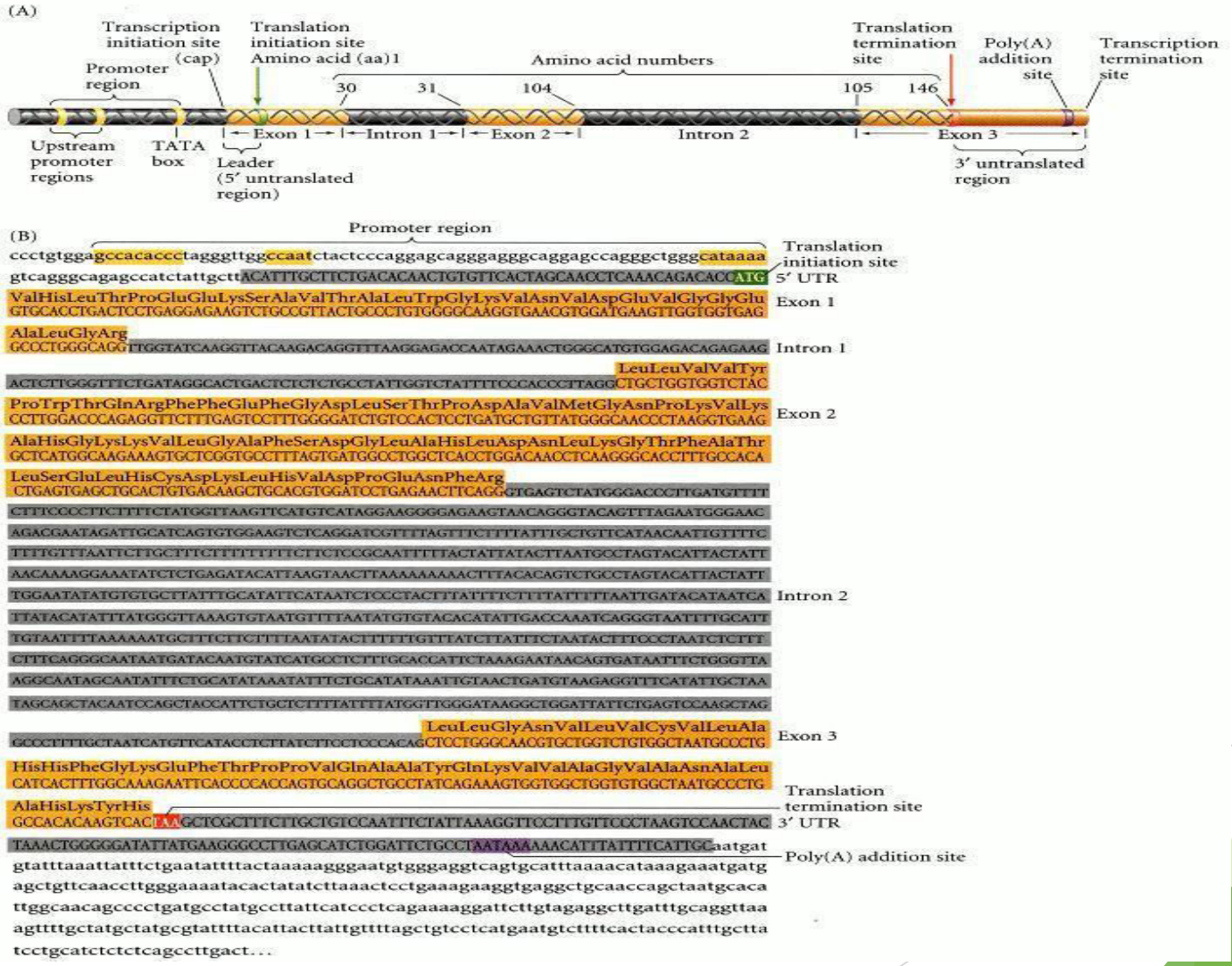
# Anatomy of the gene

**Exons:**

Coding regions of DNA.

**Introns:**

Non-coding regions.



# Anatomy of the gene

**Promoter:**

A region, which is responsible for the binding of RNA polymerase and for the subsequent initiation of transcription.

# Anatomy of the gene

**Enhancer:**

It is a DNA sequence that can activate the utilization of a promoter, controlling the efficiency and rate of transcription from that particular promoter.

The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs.

# Anatomy of the gene

**Untranslated region (UTR):**

A region that is not translated into protein.

This region includes the sequence which is needed for **polyadenylation**(addition of a poly(A) tail to a messenger RNA).

# Anatomy of the gene

The **poly(A) tail**

- Confers stability on the messenger RNA.
- Allows the mRNA to exit the nucleus.
- Permits the mRNA to be translated into protein.

# Role of Enhancers in differential gene expression

# Role of Enhancers in differential gene expression

An **enhancer** is a DNA sequence that can activate the utilization of a promoter, controlling the efficiency and rate of transcription from that particular promoter.

Enhancers can regulate the temporal and tissue-specific expression of any differentially regulated gene, but different types of genes normally have different enhancers.

# Role of Enhancers in differential gene expression

Enhancers are critical in the regulation of normal development. Over the past decade, six generalizations that emphasize their importance for differential gene expression have been made:

# Role of Enhancers in differential gene expression

- Most genes require enhancers for their transcription.
- Enhancers are the major determinant of differential transcription in space (cell type) and time.

# Role of Enhancers in differential gene expression

- A given gene can have several enhancer sites linked to it, and each enhancer can be bound by more than one transcription factor.
- The interaction between the proteins bound to the enhancer sites and the transcription initiation complex assembled at the promoter is thought to regulate transcription.

# Role of Enhancers in differential gene expression

- Enhancers are modular. It is the combination of transcription factors that causes particular genes to be transcribed.
- A gene can have several enhancer elements, each turning it on in a different set of cells.

# Role of Enhancers in differential gene expression

- Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called **silencers**.

# Transcription factors

# Transcription factors

**Transcription factors** are proteins that bind to enhancer or promoter regions and interact to activate or repress the transcription of a particular gene. Most transcription factors can bind to specific DNA sequences.

# Transcription factors

These proteins can be grouped together in families based on similarities in structure. The transcription factors within such a family share a common framework structure in their DNA-binding sites, and slight differences in the amino acids at the binding site can alter the sequence of the DNA to which the factor binds.

# Transcription factors

Transcription factors have three major domains.

1. **DNA-binding domain**
2. **Protein-protein interaction domain**
3. ***Trans*-activating domain**

# Transcription factors

**DNA-binding domain:**

It recognizes a particular DNA sequence.

**Protein-protein interaction domain:**

It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.

# Transcription factors

***Trans*-activating domain:**

It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually, this *trans*-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.

# Transcription factors

Examples,

***MITF:***

*(*Microphthalmia-Associated Transcription Factor. The microphthalmia (MITF) protein is necessary for the production of pigment cells and their pigments.

# Transcription factors

***Pax6:***

Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye, nervous system, and pancreas development, contains two potential DNA-binding domains.

# Transcription factors

**Pdx1:**

Pancreatic And Duodenal Homeobox 1 gene.

It is specific for the pancreatic region of the endoderm.

# Fertilization

# Fertilization

Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

# Fertilization

Fertilization accomplishes two separate processes:

- Sex (the combining of genes derived from the two parents) and
- Reproduction (the creation of new organisms)

# Fertilization

Thus, the first function of fertilization is to transmit genes from parent to offspring, and the second is to initiate in the egg cytoplasm those reactions that permit development to proceed.

# Fertilization

Although the details of fertilization vary from species to species, conception generally consists of four major events:

- Contact and recognition between sperm and egg. In most cases, this ensures that the sperm and egg are of the same species.

# Fertilization

- Regulation of sperm entry into the egg. Only one sperm can ultimately fertilize the egg.
- Fusion of the genetic material of sperm and egg.
- Activation of egg metabolism to start development.

# Structure of the Sperm

# Structure of the Sperm

**A.V Leeuwenhoek**, the Dutch microscopist who co-discovered sperm in 1678, first believed them to be parasitic animals living within the semen (hence the term *spermatozoa*, meaning "sperm animals").

# Structure of the Sperm

Leeuwenhoek (1685) wrote that sperm were seeds (both *sperma* and *semen* mean "seed") and that the female merely provided the nutrient soil in which the seeds were planted.

# Structure of the Sperm

The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by **Lazzaro Spallanzani** in the late 1700s. Spallanzani demonstrated that filtered toad semen devoid of sperm would not fertilize eggs.

# Structure of the Sperm

In 1824, **J. L. Prevost and J. B. Dumas** claimed that sperm were not parasites, but rather the active agents of fertilization.

“They proposed that the sperm entered the egg and contributed materially to the next generation”.

# Structure of the Sperm

In 1876 **Oscar Hertwig and Herman Fol** independently demonstrated sperm entry into the egg and the union of the two cells' nuclei.

“Fertilization was at last recognized as the union of sperm and egg, and the union of sea urchin gametes remains one of the best-studied examples of fertilization”.

# Structure of the Sperm II

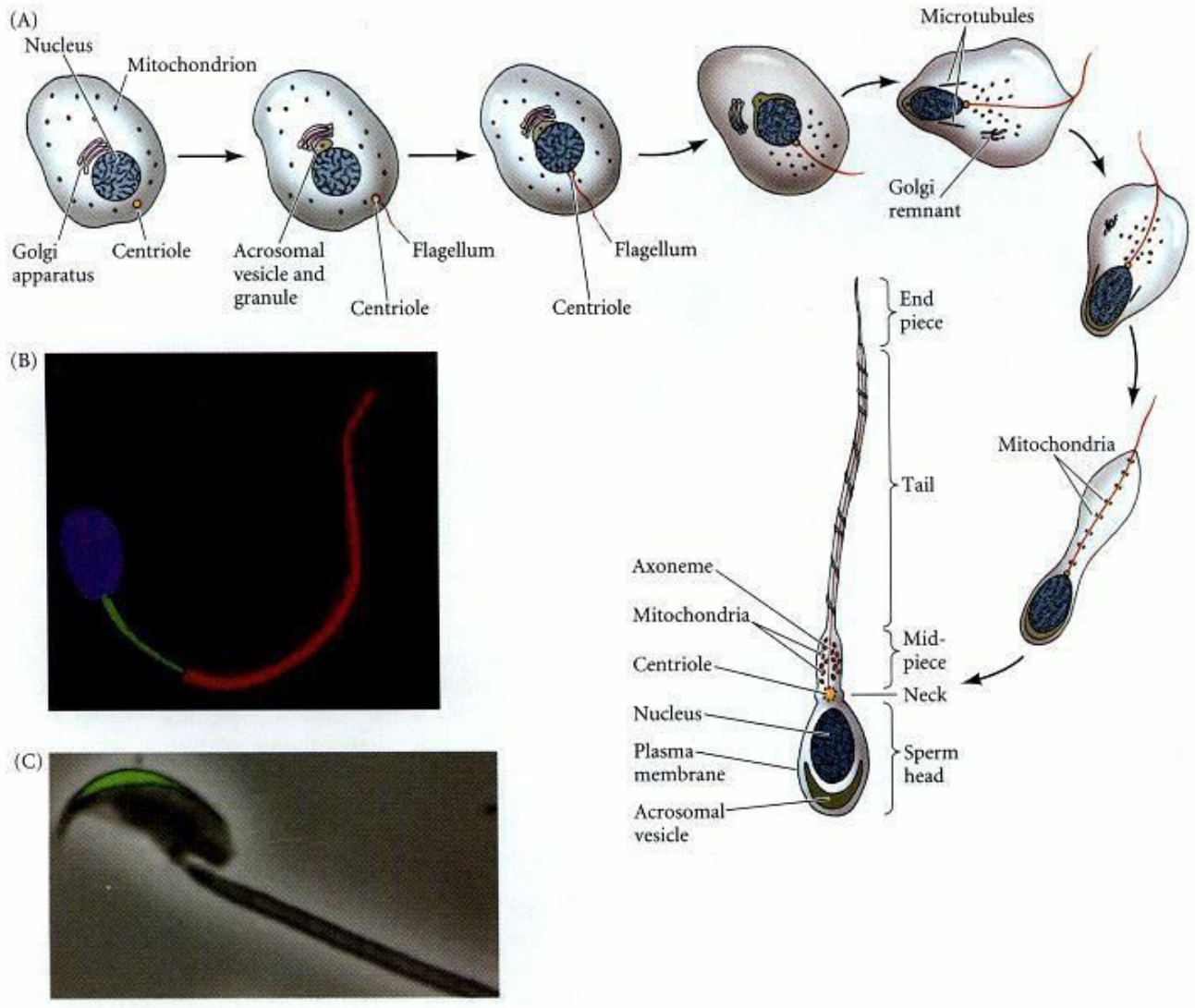
# Structure of the Sperm II

Each sperm consists of,

- A haploid nucleus,
- A propulsion system to move the nucleus, and
- A sac of enzymes that enable the nucleus to enter the egg.

# Structure of the Sperm II

During the course of sperm maturation, the haploid nucleus becomes very streamlined, and its DNA becomes tightly compressed



# Structure of the Sperm II

In front of this compressed haploid nucleus lies the **acrosomal vesicle**, or **acrosome**, which is derived from the Golgi apparatus and contains enzymes that digest proteins and complex sugars; thus, it can be considered a modified secretory vesicle.

# Structure of the Sperm II

These stored enzymes are used to lyse the outer coverings of the egg. In many species, such as sea urchins, a region of globular actin molecules lies between the nucleus and the acrosomal vesicle.

These proteins are used to extend a fingerlike **acrosomal process** from the sperm during the early stages of fertilization.

# Structure of the Sperm II

In sea urchins and several other species, recognition between sperm and egg involves molecules on the acrosomal process. Together, the acrosome and nucleus constitute the head of the sperm.

# Structure of the Sperm II

The means by which sperm are propelled vary according to how the species has adapted to environmental conditions. In some species (such as the **parasitic roundworm** ***Ascaris***), the sperm travel by the amoeboid motion of **lamellipodial extensions** of the cell membrane.

# Structure of the Sperm II

In most species, however, each sperm is able to travel long distances by whipping its **flagellum.** Flagella are complex structures. The major motor portion of the flagellum is called the **axoneme**. It is formed by microtubules emanating from the centriole at the base of the sperm nucleus.

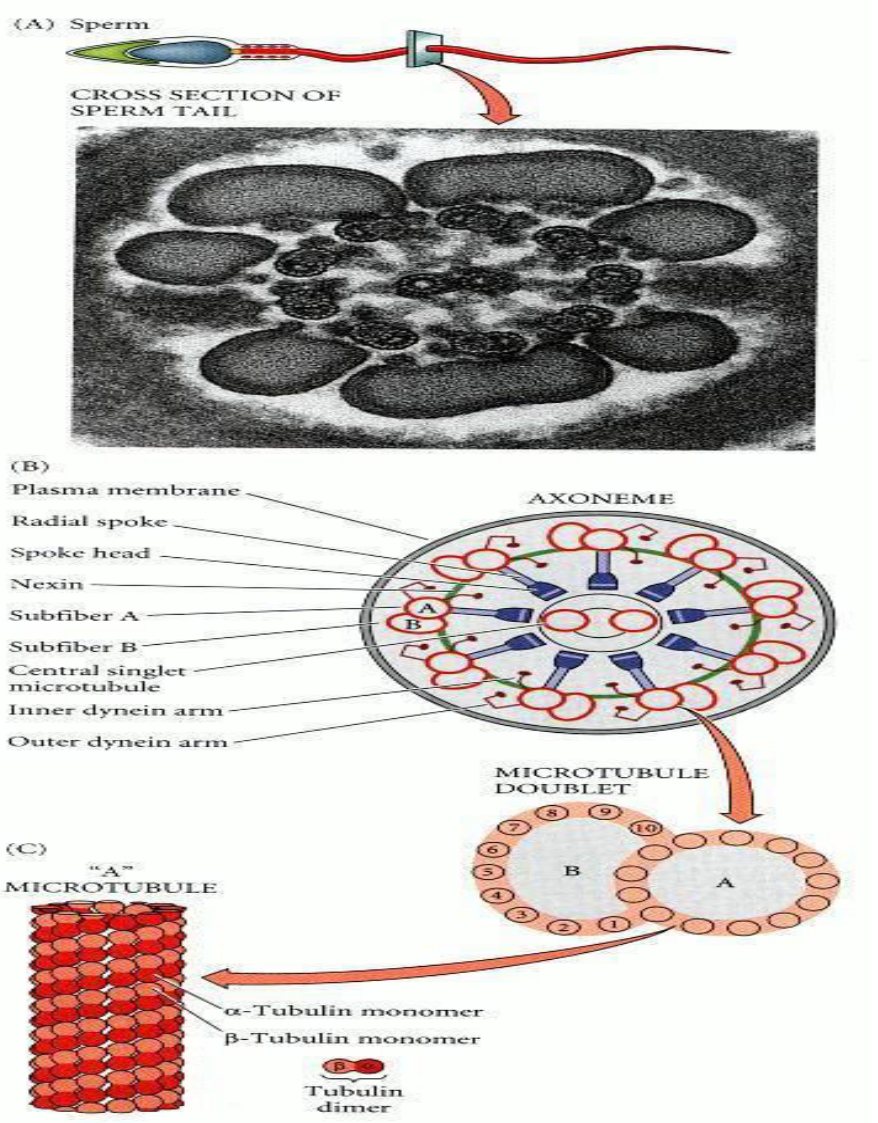
# Structure of the Sperm II

The core of the axoneme consists of two central microtubules surrounded by a row of nine doublet microtubules.

The force for sperm propulsion is provided by **dynein**, a protein that is attached to the microtubules.

# Structure of the Sperm II

Dynein hydrolyzes molecules of ATP and can convert the released chemical energy into the mechanical energy that propels the sperm.



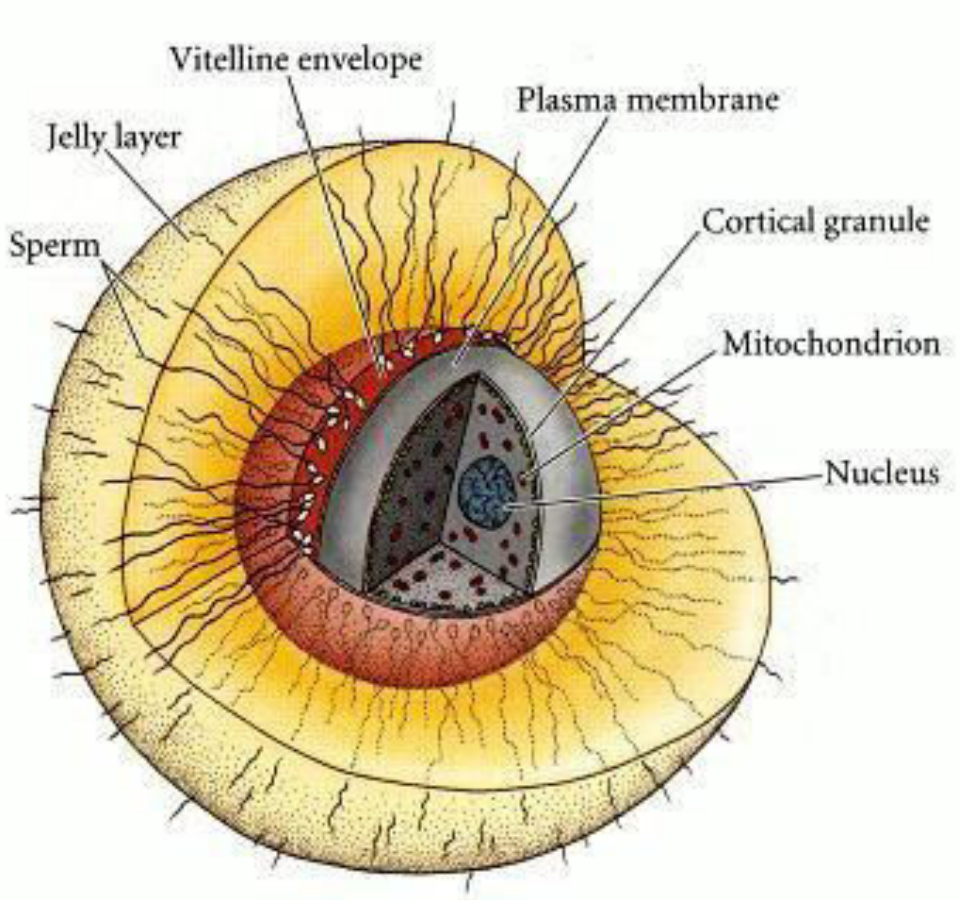
# Structure of the Sperm II

The sperm released during ejaculation are able to move, yet they do not have the capacity to bind to and fertilize an egg. These final stages of sperm maturation (**called capacitation**) do not occur until the sperm has been inside the female reproductive tract for a certain period of time.

# Structure of Egg

# Structure of Egg

The volume of a sea urchin egg is about 200 picoliters (**2** $\times$**10**$^{-4}$ mm3, more than 10,000 times the volume of the sperm)



# Structure of Egg

The egg also has a remarkable cytoplasmic storehouse that it has accumulated during its maturation. This cytoplasmic storehouse includes the following,

## Proteins:

The early embryonic cells need a supply of energy and amino acids which is provided by yolk proteins in the egg. Many of the yolk proteins are made in other organs (liver, fat body) and travel through the maternal blood to the egg.

# Structure of Egg

**Ribosomes and tRNA:**

Protein synthesis is accomplished by ribosomes and tRNA, which exist in the egg. The developing egg has special mechanisms to synthesize ribosomes, and certain amphibian oocytes produce as many as $\mathbf{10^{12}}$ ribosomes.

# Structure of Egg

**Messenger RNA:**

The instructions for proteins made during early development are already packaged in the oocyte. It is estimated that the eggs of sea urchins contain 25,000 to 50,000 different types of mRNA.

# Structure of Egg

**Morphogenetic factors:**

Molecules that direct the differentiation of cells into certain cell types are localized in different regions of the egg and become segregated into different cells during cleavage.

# Structure of Egg

**Protective chemicals:**

Many eggs contain ultraviolet filters and DNA repair enzymes that protect them from sunlight; some eggs contain molecules that potential predators find distasteful; and the yolk of bird eggs even contains antibodies.

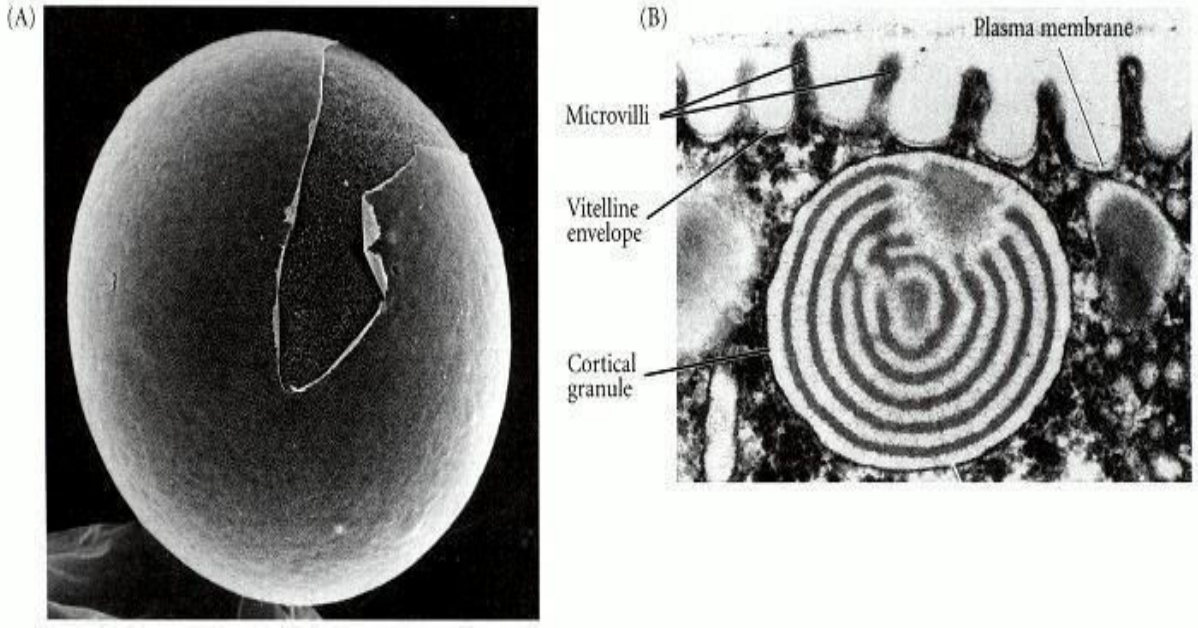
# Structure of Egg

Many types of eggs have glycoprotein meshwork called **egg jelly** outside the vitelline envelope which is used either to attract or to activate sperm. The egg, then, is a cell specialized for receiving sperm and initiating development.

# Structure of Egg II

# Structure of Egg II

Enclosing the cytoplasm is the egg **plasma membrane** that regulates the flow of certain ions during fertilization and must be capable of fusing with the sperm plasma membrane.

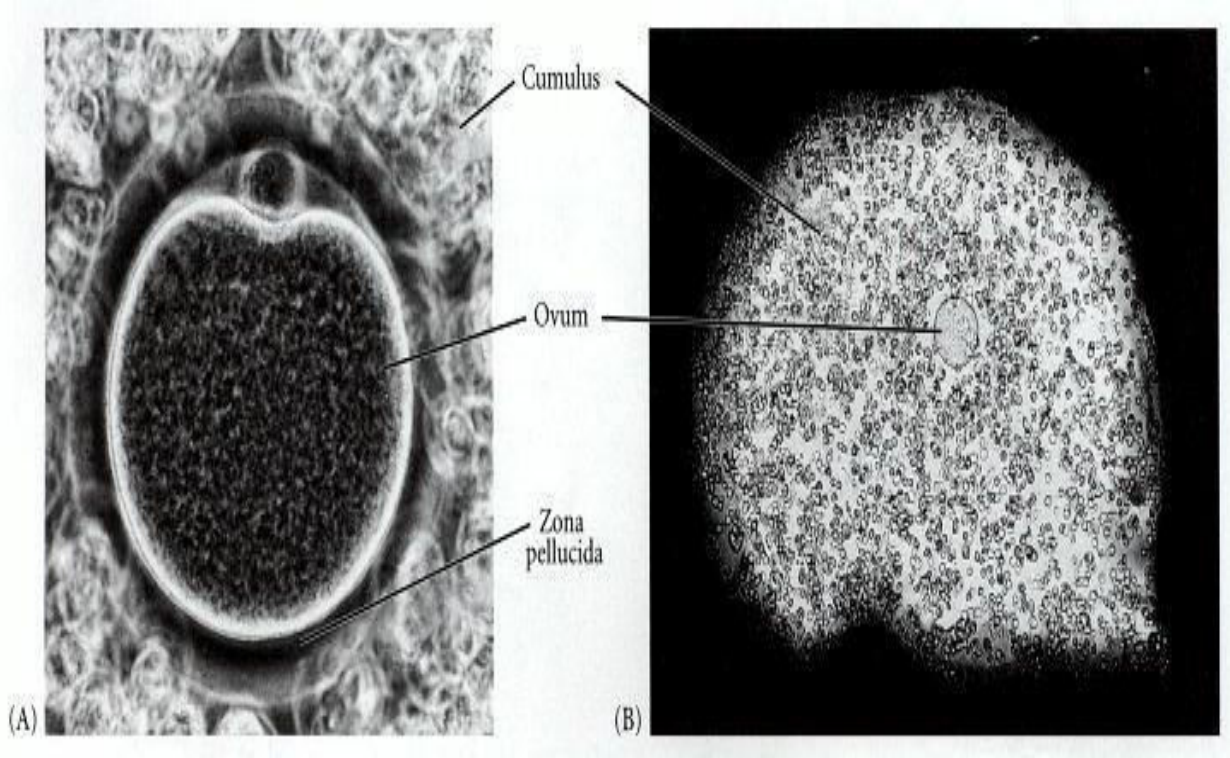


# Structure of Egg II

Outside the plasma membrane is the **vitelline envelope**, which forms a fibrous mat around the egg. This envelope contains at least eight different glycoproteins and is often involved in sperm-egg recognition.

# Structure of Egg II

The vitelline envelope is essential for the species-specific binding of sperm. In mammals, the vitelline envelope is a separate and thick extracellular matrix called the **zona pellucida**.



# Structure of Egg II

The mammalian egg is also surrounded by a layer of cells called the **cumulus.**

The innermost layer of cumulus cells, immediately adjacent to the zona pellucida, is called the **corona radiata.**

# Structure of Egg II

Lying immediately beneath the plasma membrane of the egg is a thin shell (about 5 μm) of gel-like cytoplasm called the **cortex**. The cytoplasm in this region is stiffer than the internal cytoplasm and contains high concentrations of globular actin molecules.

# Structure of Egg II

During fertilization, these actin molecules polymerize to form long cables of actin known as **microfilaments**.

Microfilaments are necessary for cell division, and they also are used to extend the egg surface into small projections called **microvilli**, which may aid sperm entry into the cell.

# Structure of Egg II

**Cortical granules** are membrane-bound structures containing proteolytic enzymes.

However, whereas each sperm contains one acrosomal vesicle, each sea urchin egg contains approximately 15,000 cortical granules.

# Recognition of Egg and Sperm

# Recognition of Egg and Sperm

**The interaction of sperm and egg generally proceeds according to five basic step:**

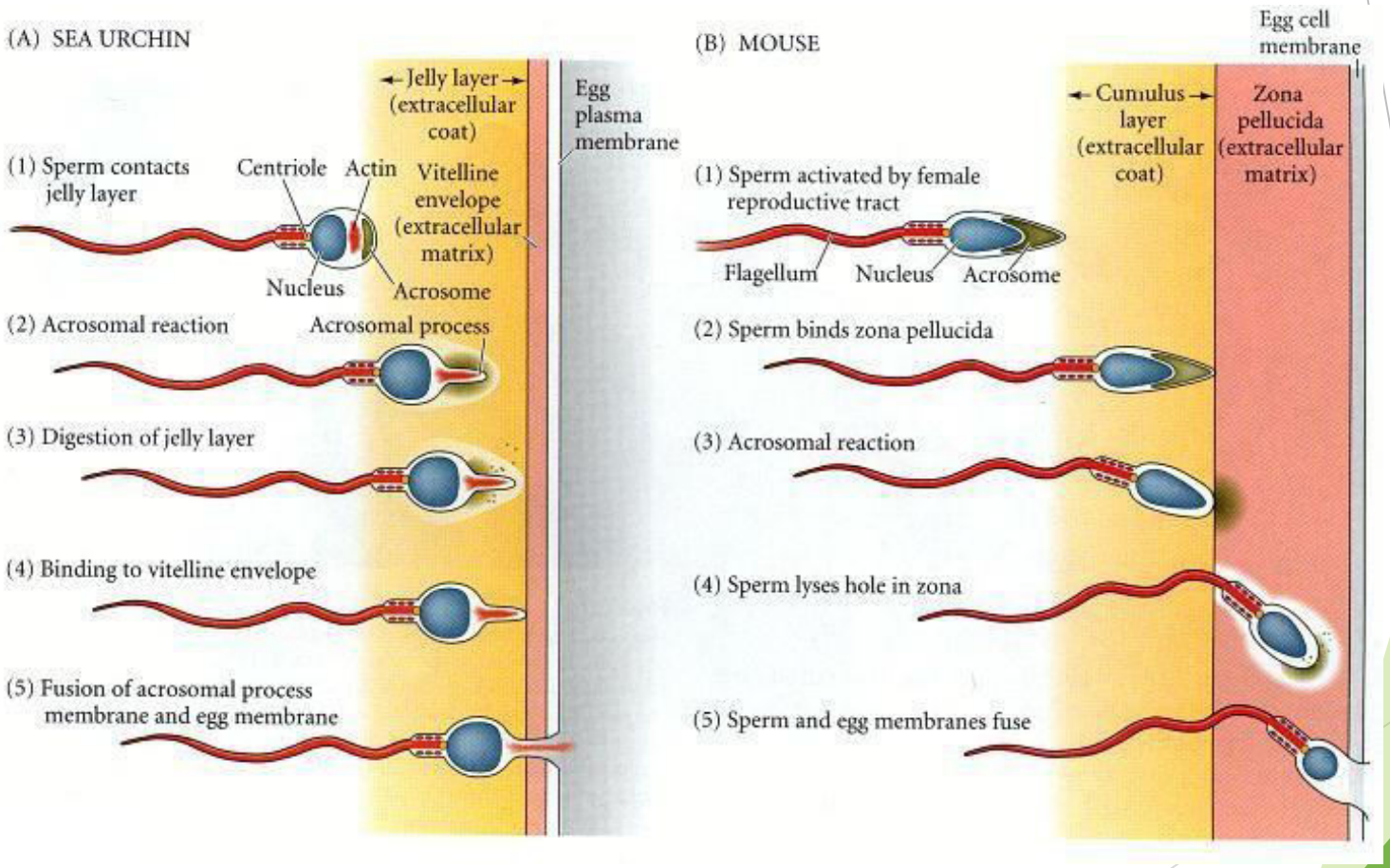
- The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg.
- The exocytosis of the acrosomal vesicle to release its enzymes.

# Recognition of Egg and Sperm

- The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg.
- The passing of the sperm through this extracellular envelope.

# Recognition of Egg and Sperm

- Fusion of egg and sperm cell plasma membranes.



# Recognition of Egg and Sperm

In many species, the meeting of sperm and egg is not a simple matter. Many marine organisms release their gametes into the environment. That environment may be as small as a tide pool or as large as an ocean.

# Recognition of Egg and Sperm

These organisms are faced with two problems:

- How can sperm and eggs meet in such a dilute concentration?
- How can sperm be prevented from trying to fertilize eggs of another species?

# Recognition of Egg and Sperm

Two major mechanisms have evolved to solve these problems:

- Species specific attraction of sperm.
- Species-specific sperm activation.

# Sperm attraction

# Sperm attraction

Species-specific sperm attraction has been documented in numerous species, including cnidarians, molluscs, echinoderms, and urochordates. In many species, sperm are attracted toward eggs of their species by **chemotaxis**, that is, by following a gradient of a chemical secreted by the egg.

# Sperm attraction

In 1978, Miller demonstrated that the eggs of the cnidarian ***Orthopyxis caliculata*** not only secrete a chemotactic factor but also regulate the timing of its release.

# Sperm attraction

Miller found that when sperm were added to oocytes that had not yet completed their second meiotic division, there was no attraction of sperm to eggs.

# Sperm attraction

However, after the second meiotic division was finished and the eggs were ready to be fertilized, the sperm migrated toward them.

Thus, these oocytes control not only the type of sperm they attract, but also the time at which they attract them.

# Sperm attraction

The mechanisms of **chemotaxis** differ among species. One chemotactic molecule, a 14-amino acid peptide called **resact,** has been isolated from the egg jelly of the sea urchin *Arbacia punctulata*.

# Sperm attraction

**Resact** diffuses readily in seawater and has a profound effect at very low concentrations when added to a suspension of *Arbacia* sperm.

**Resact** is specific for ***A. punctulata*** and does not attract sperm of other species.

# Sperm attraction

*A. punctulata* sperm have receptors in their plasma membranes that bind resact and can swim up a concentration gradient of this compound until they reach the egg.

# Sperm attraction

Resact also acts as a **sperm-activating peptide**.

**The sperm receptor for Resact** is a trans membrane protein, and when it binds Resact on the extracellular side, a conformational change on the cytoplasmic side activates the receptor's enzymatic activity.

# Sperm attraction

This activates the mitochondrial ATP-generating apparatus as well as the dynein ATPase that stimulates flagellar movement in the sperm.

# The acrosomal reaction in sea urchins

# The acrosomal reaction in sea urchins

In most marine invertebrates, the acrosomal reaction has two components:

- The fusion of the acrosomal vesicle with the sperm plasma membrane (an exocytosis that results in the release of the contents of the acrosomal vesicle).
- The extension of the acrosomal process.

# The acrosomal reaction in sea urchins

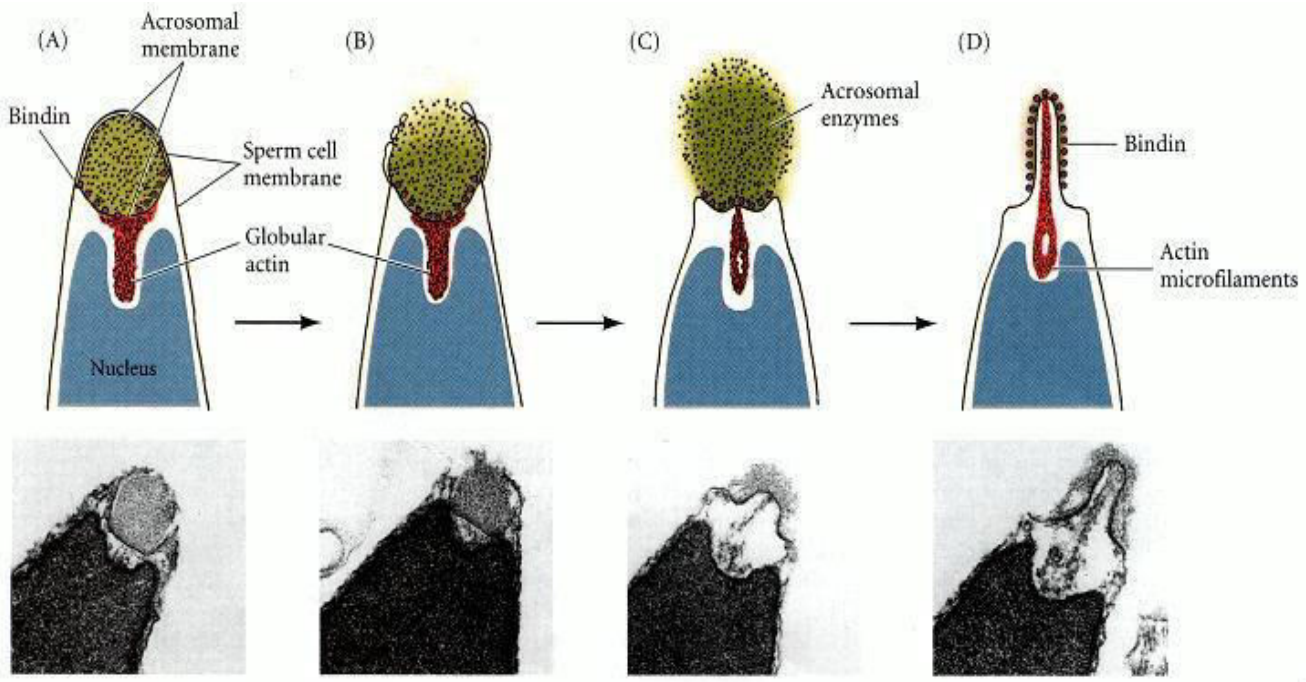
The acrosomal reaction in sea urchins is initiated by contact of the sperm with the egg jelly. Contact with egg jelly causes the exocytosis of the sperm's acrosomal vesicle and the release of proteolytic enzymes that can digest a path through the jelly coat to the egg surface.

# The acrosomal reaction in sea urchins

In sea urchins, the acrosomal reaction is thought to be initiated by a fucose-containing polysaccharide in the egg jelly that binds to the sperm and allows calcium to enter into the sperm head.

# The acrosomal reaction in sea urchins

The exocytosis of the acrosomal vesicle is caused by the calcium-mediated fusion of the acrosomal membrane with the adjacent sperm plasma membrane.

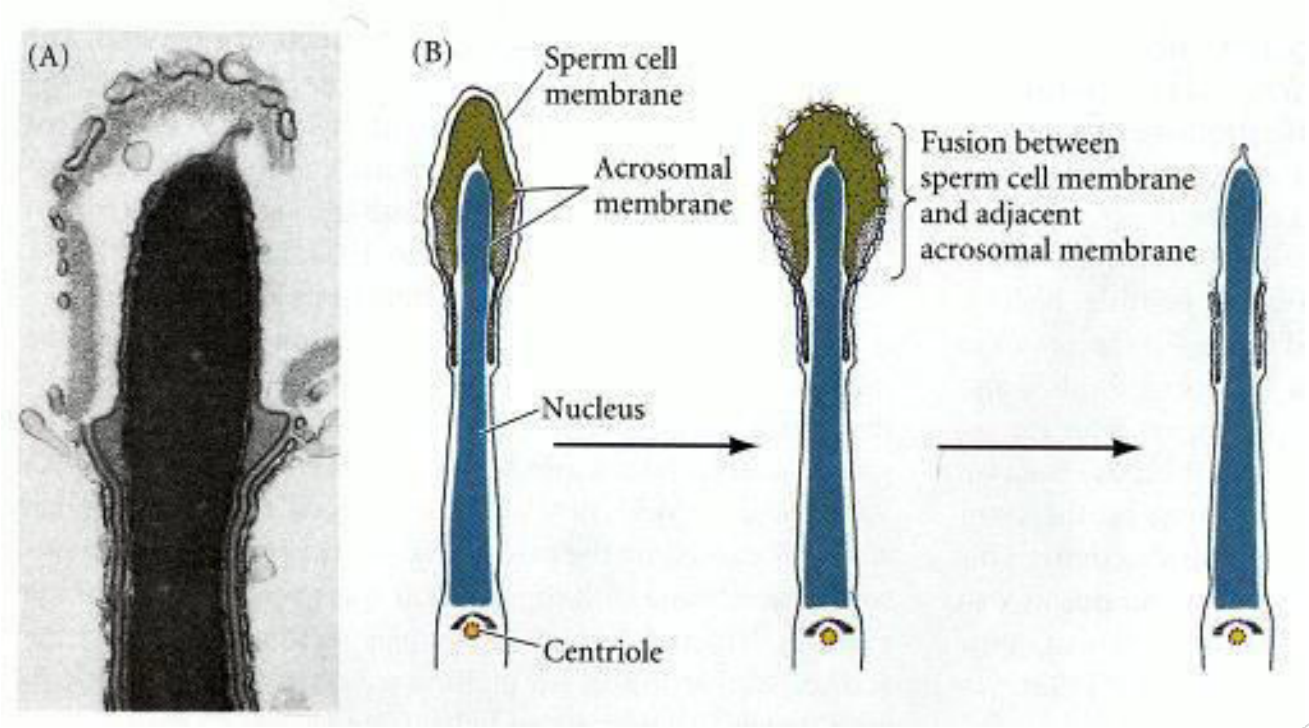


# The acrosomal reaction in sea urchins

The second part of the acrosomal reaction involves the extension of the acrosomal process. This protrusion arises through the polymerization of globular actin molecules into actin filaments.

# The acrosomal reaction in sea urchins

The egg jelly factors that initiate the acrosomal reaction in sea urchins are often highly specific to each species.



# Species-specific recognition in sea urchins

# Species-specific recognition in sea urchins

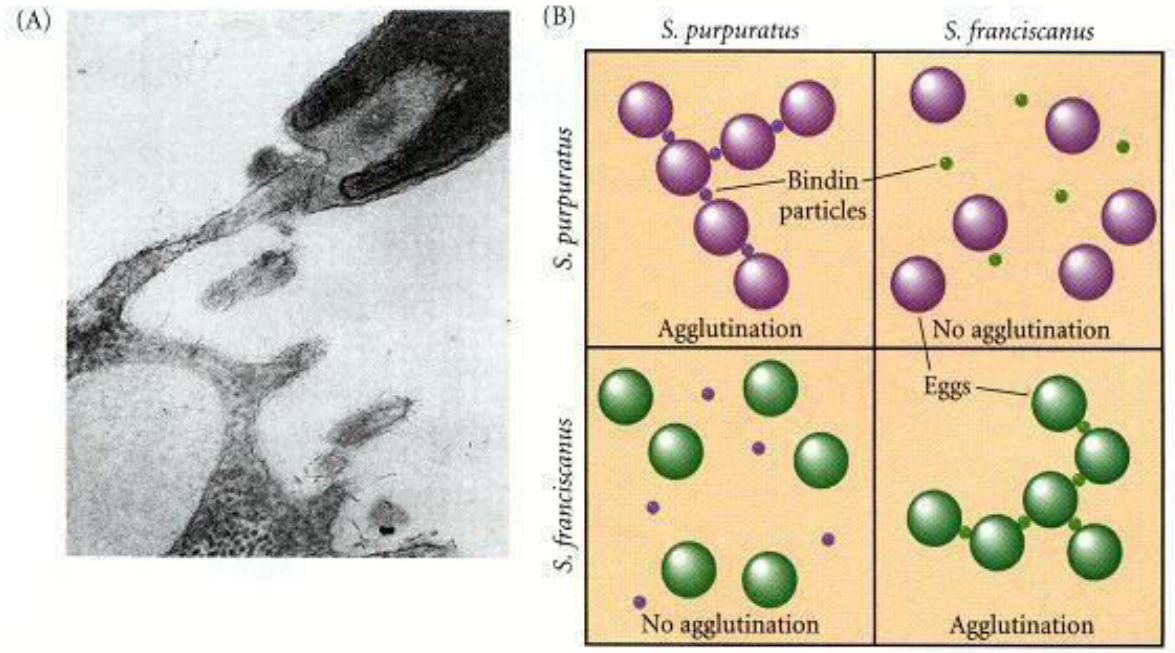
Once the sea urchin sperm has penetrated the egg jelly, the acrosomal process of the sperm contacts the surface of the egg . A major species-specific recognition step occurs at this point.

# Species-specific recognition in sea urchins

The acrosomal protein mediating this recognition is called **bindin**. In 1977, Vacquier and co-workers isolated this nonsoluble 30,500-Da (Dalton) protein from the acrosome of *Strongylocentrotus purpuratus* and found it to be capable of binding to dejellied eggs of the same species.

# Species-specific recognition in sea urchins

The interaction bindin with eggs is relatively species-specific. **Bindin** isolated from the acrosomes of *S. purpuratus* binds to its own dejellied eggs, but not to those of *Arbacia punctulata.*

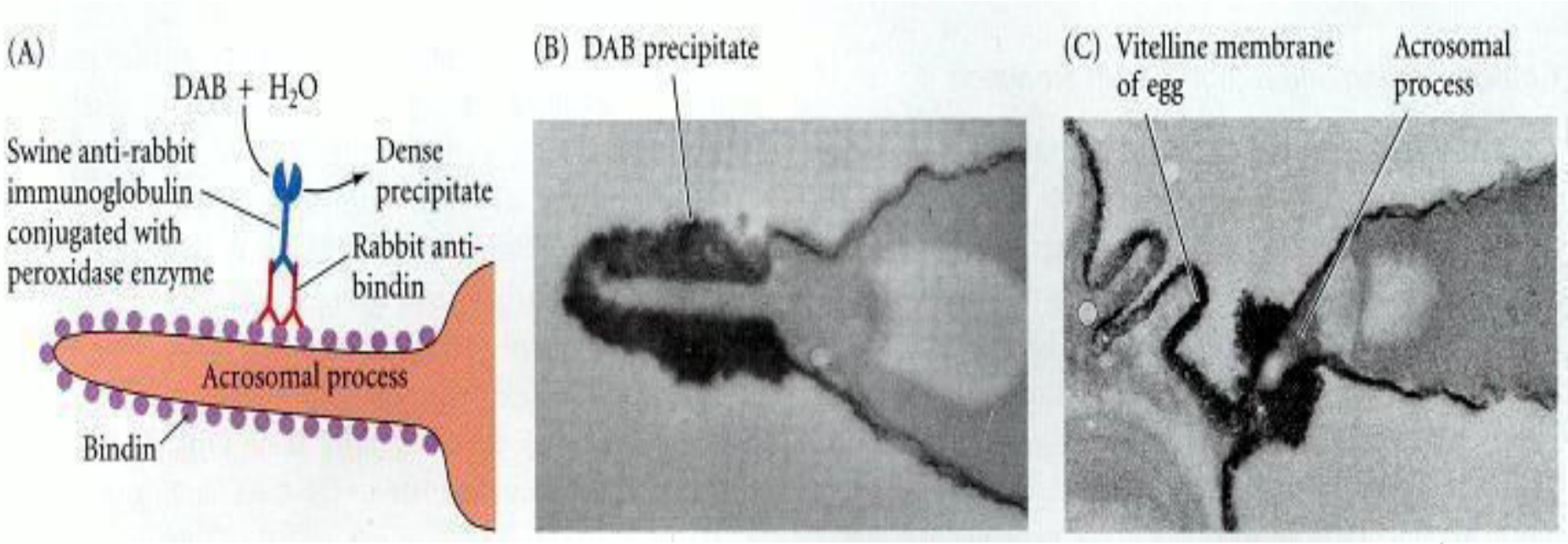


# Species-specific recognition in sea urchins

Using immunological techniques, **Moy and Vacquier** demonstrated that **bindin** is located specifically on the acrosomal process exactly where it should be for sperm-egg recognition.

# Species-specific recognition in sea urchins

As seen in Figure, sperm binding does not occur over the entire egg surface.



# Species-specific recognition in sea urchins

Thus, species-specific recognition of sea urchin gametes occurs at the levels of:

- **Sperm attraction**
- **Sperm activation**
- **Sperm adhesion to the egg surface.**

# Gamete binding and recognition in mammals

Effort By: AARISH ALI

# Gamete binding and recognition in mammals

**The zona pellucida is a glycoprotein matrix**, which is synthesized and secreted by the growing oocyte.

In mammals the zona pellucida plays a role analogous to that of the vitelline envelope in invertebrates.

# Gamete binding and recognition in mammals

**The zona pellucida** plays two major roles during fertilization:

- It binds the sperm.
- It initiates the acrosomal reaction after the sperm is bound.

# Gamete binding and recognition in mammals

The binding of sperm to the zona is relatively, but not absolutely, species-specific.

Species specific gamete recognition is not a major problem when fertilization occurs internally.

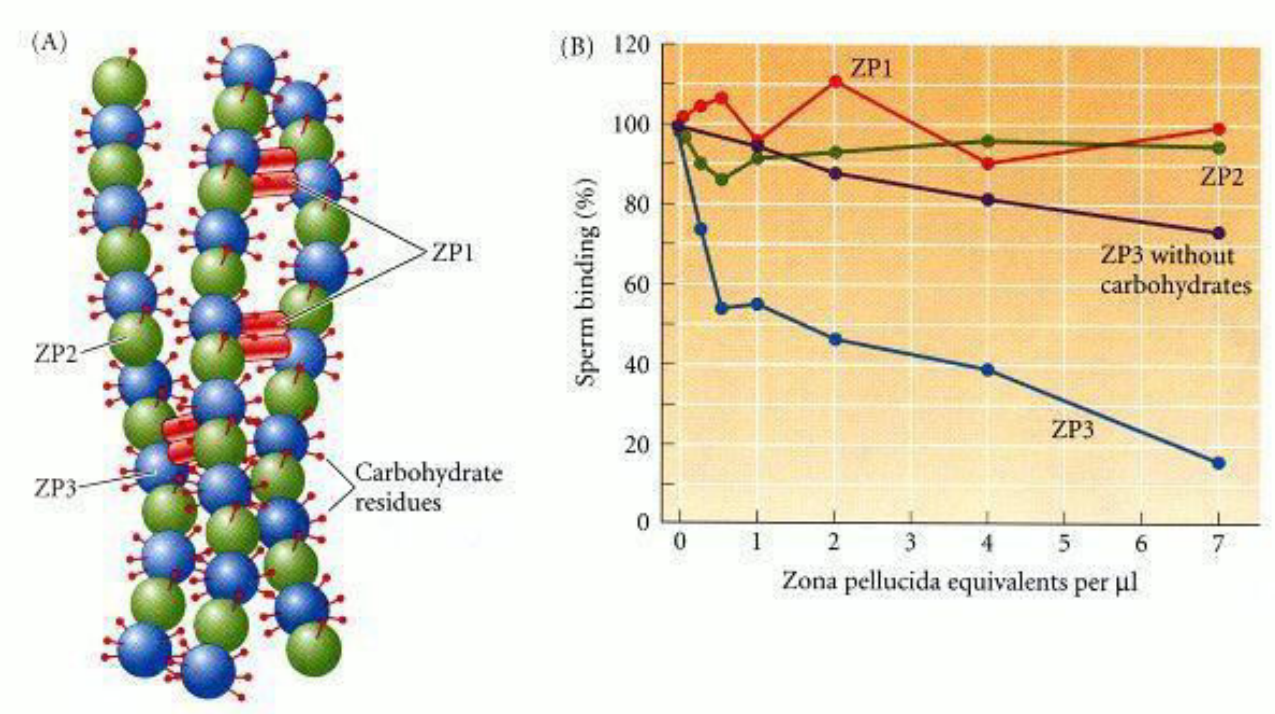
# Gamete binding and recognition in mammals

Bleil and Wassarman (1988) isolated an 83-kDa glycoprotein, **ZP3**, from the mouse zona that was the active site for sperm binding.

The other two zona glycoproteins they found, **ZP1** and **ZP2**, failed to compete for sperm binding.

# Gamete binding and recognition in mammals

Thus, **ZP3** is the specific glycoprotein in the mouse zona pellucida to which the sperm bind.



# Gamete binding and recognition in mammals

**ZP3** also initiates the acrosomal reaction after sperm have bound to it. The mouse sperm can thereby concentrate its proteolytic enzymes directly at the point of attachment at the zona pellucida.

# Induction of the mammalian acrosomal reaction

# Induction of the mammalian acrosomal reaction

Unlike the sea urchin acrosomal reaction, the acrosomal reaction in mammals occurs only after the sperm has bound to the zona pellucida.

# Induction of the mammalian acrosomal reaction

The opening of the calcium channels involve the receptor's activating a cation channel (for sodium, potassium, or calcium), which would change the resting potential of the sperm plasma membrane.

The calcium channels in the membrane would be sensitive to this change in membrane potential, allowing calcium to enter the sperm.

# Induction of the mammalian acrosomal reaction

The difference between the acrosomal reaction in sea urchins and mammals may be due to the thickness of the extracellular envelopes surrounding the egg. In the sea urchin, the vitelline envelope is very thin and porous.

# Induction of the mammalian acrosomal reaction

Once a sperm has bound there, it is very close to the egg plasma membrane, and, indeed, the bindin receptor may extend through the vitelline envelope. In mammals, however, the zona pellucida is a very thick matrix, so the sperm is far removed from the egg.

# Induction of the mammalian acrosomal reaction

By undergoing the acrosomal reaction directly on the zona, the sperm is able to concentrate its proteolytic enzymes to lyse a hole in this envelope. Indeed, sperm that undergo the acrosomal reaction before they reach the zona pellucida are unable to penetrate it.

# Induction of the mammalian acrosomal reaction

It appears that the acrosome-reacted sperm transfer their binding from ZP3 to the adjacent ZP2 molecules. After a mouse sperm has entered the egg, the egg cortical granules release their contents.

# Induction of the mammalian acrosomal reaction

One of the proteins released by these granules is a protease that specifically alters **ZP2** (Moller and Wassarman 1989).

This inhibits other acrosome-reacted sperm from moving closer toward the egg.

# Capacitation

# Capacitation

The set of physiological changes that allow the sperm to be competent to fertilize the egg is called **capacitation**. The requirement for capacitation varies from species to species.

# Capacitation

**Wilcox and colleagues (1995)** found that nearly all human pregnancies result from sexual intercourse during a **6-day period** ending on the day of ovulation.

This means that the fertilizing sperm could have taken as long as 6 days to make the journey.

# Capacitation

The molecular changes that account for capacitation are,

- First, the fluidity of the sperm plasma membrane is altered by the removal of cholesterol by albumin proteins found in the female reproductive tract.

# Capacitation

- Second, particular proteins or carbohydrates on the sperm surface are lost during capacitation.

It is possible that these compounds block the recognition sites for the proteins that bind to the zona pellucida.

# Capacitation

- Third, the membrane potential of the sperm becomes more negative as potassium ions leave the sperm.

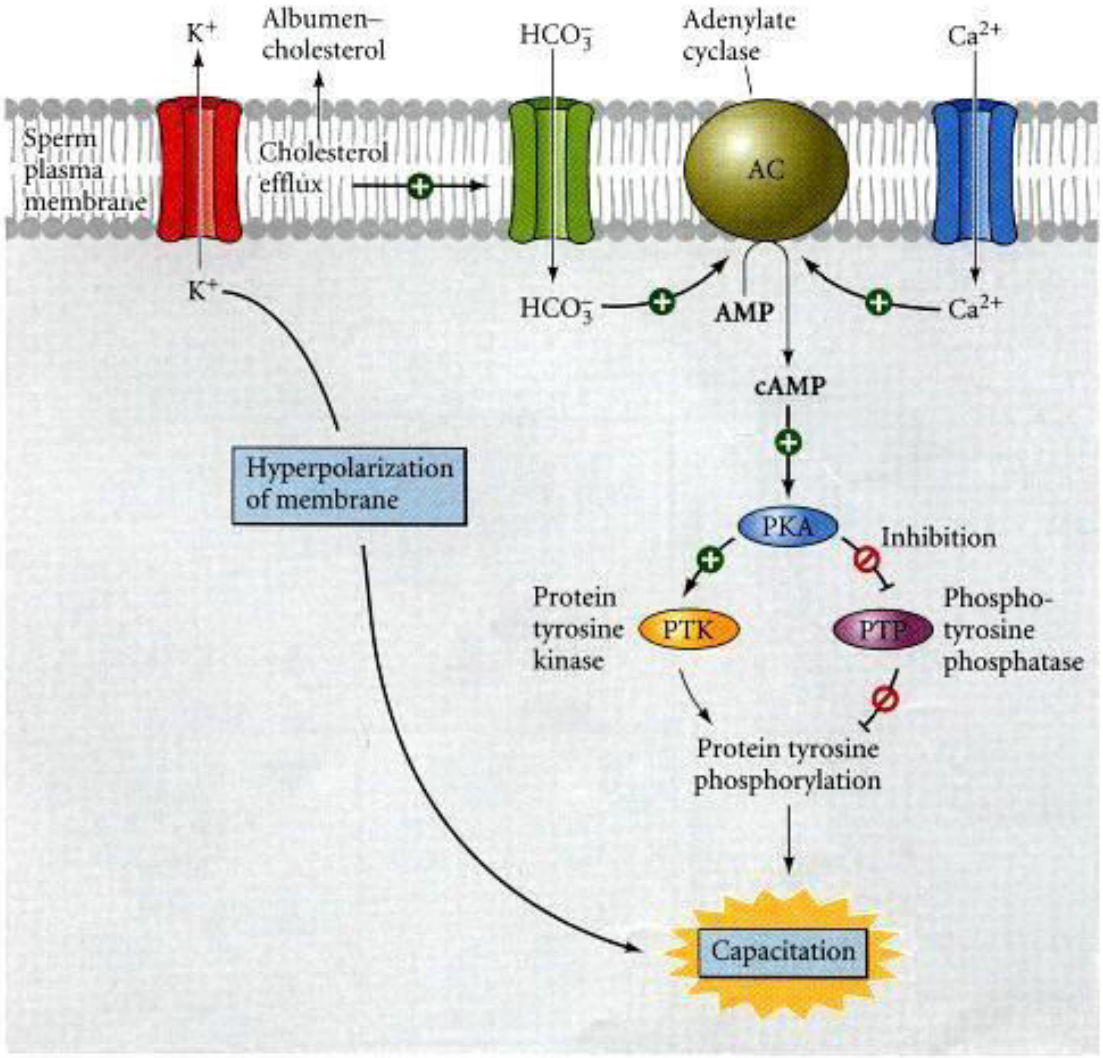
This change in membrane potential may allow calcium channels to be opened and permit calcium to enter the sperm.

# Capacitation

Calcium and bicarbonate ions may be critical in activating **cAMP** (Cyclic adenosine monophosphate) production and in facilitating the membrane fusion events of the acrosomal reaction.

# Capacitation

- Fourth, protein phosphorylation occurs.



# Gamete Fusion

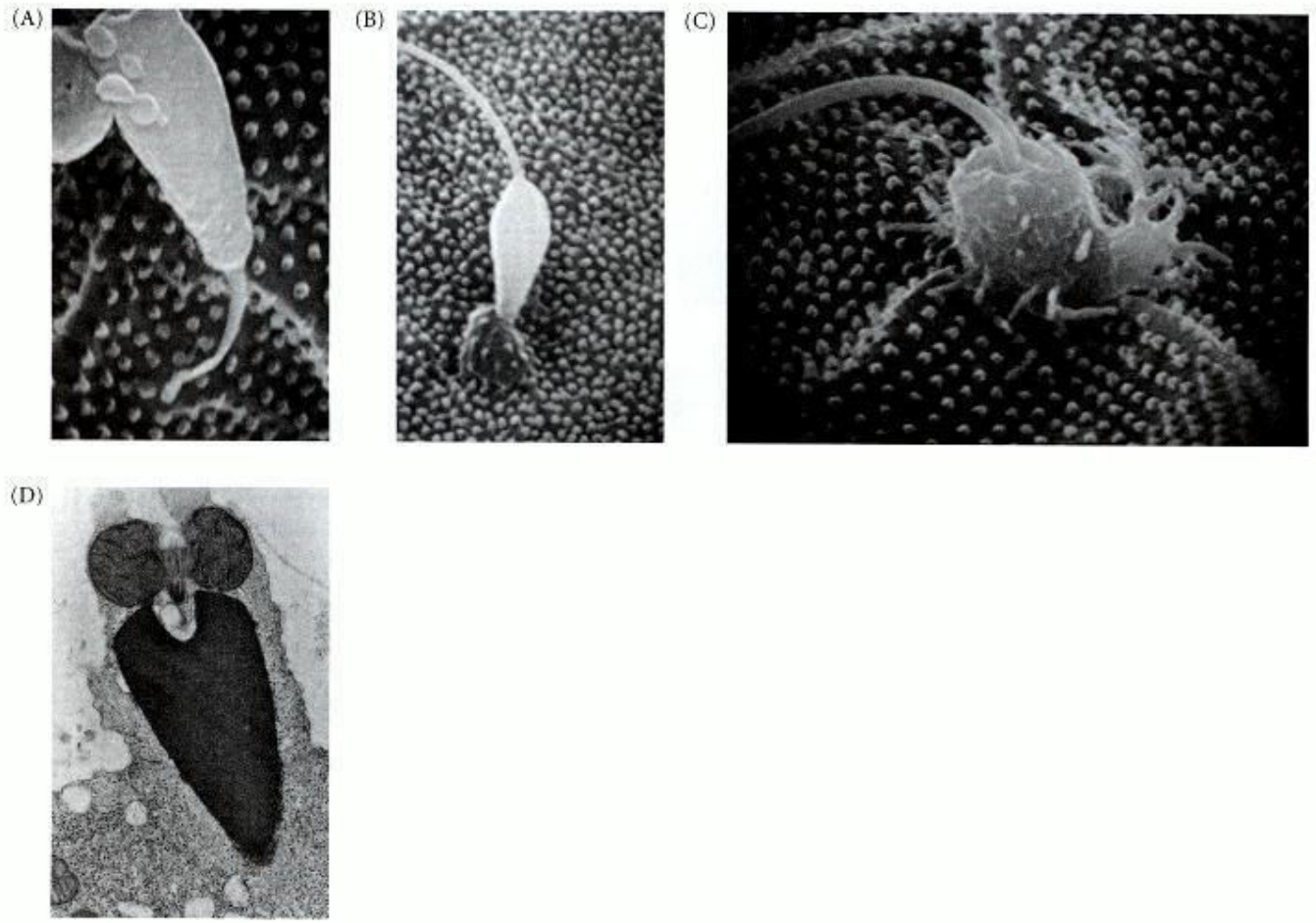
# Gamete Fusion

Recognition of sperm by the vitelline envelope or zona pellucida is followed by the lysis of that portion of the envelope or zona in the region of the sperm head by the acrosomal enzymes.

This lysis is followed by the fusion of the sperm plasma membrane with the plasma membrane of the egg.

# Gamete Fusion

The entry of a sperm into a sea urchin egg is illustrated in this Figure.



# Gamete Fusion

The sperm and egg plasma membranes then join together, and material from the sperm membrane can later be found on the egg membrane. The sperm nucleus and tail pass through the resulting cytoplasmic bridge, which is widened by the actin polymerization.

# Gamete Fusion

In the sea urchin, all regions of the egg plasma membrane are capable of fusing with sperm. In several other species, certain regions of the membrane are specialized for sperm recognition and fusion. Fusion is an active process, often mediated by specific "**fusogenic" proteins**.

# Gamete Fusion

In mammals, the **fertilin** proteins in the sperm plasma membrane are essential for sperm membrane-egg membrane fusion.

Mouse fertilin is localized to the posterior plasma membrane of the sperm head.

# Gamete Fusion

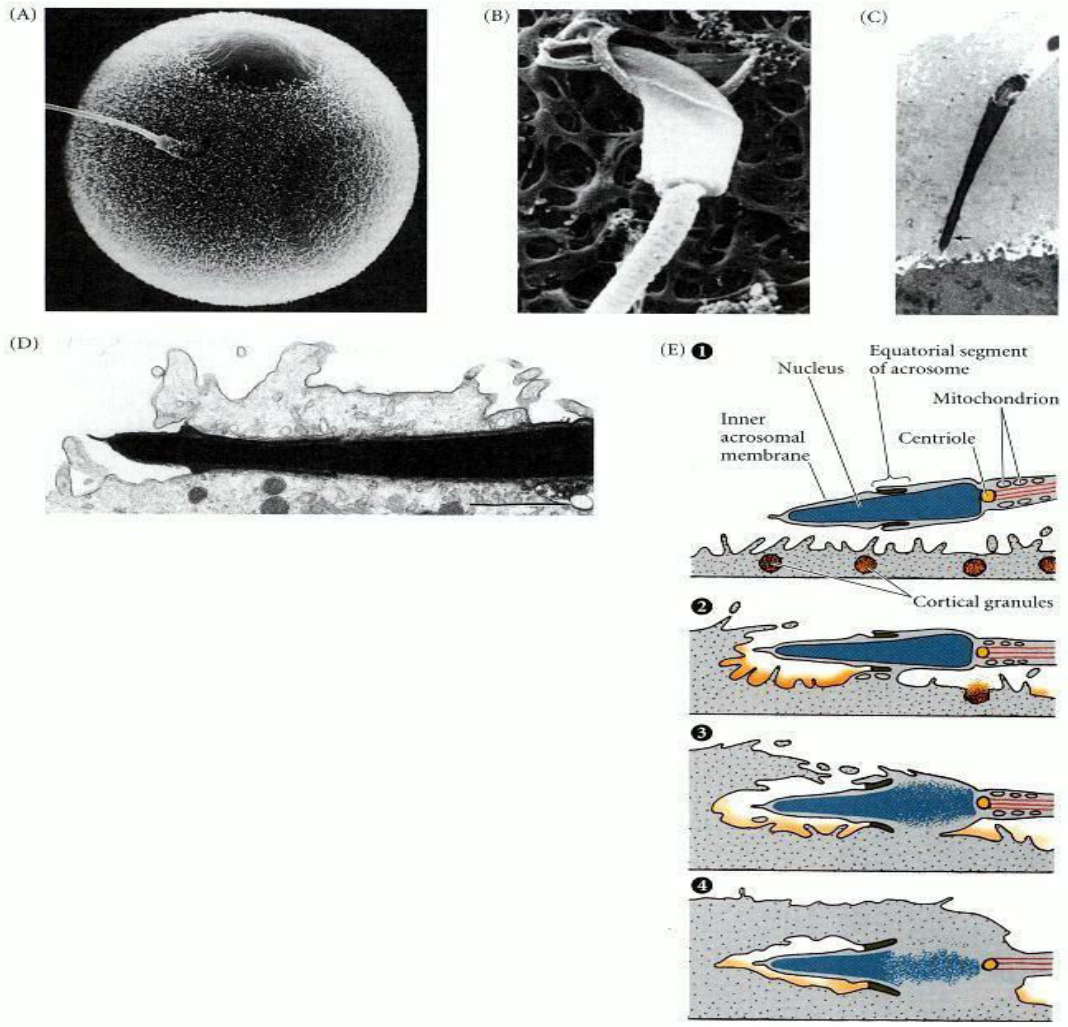
It adheres the sperm to the egg by binding to the **6β1 integrin protein** on the egg plasma membrane. Moreover, like sea urchin bindin (to which it is not structurally related), fertilin has a hydrophobic region that could potentially mediate the union of the two membranes.

# Gamete Fusion

Thus, **fertilin** appears to bind the sperm plasma membrane to the egg plasma membrane and then to fuse the two of them together.

# Gamete Fusion

When the membranes are fused, the sperm nucleus, mitochondria, centriole, and flagellum can enter the egg.



# The prevention of polyspermy

# The prevention of polyspermy

In normal **monospermy**, in which only one sperm enters the egg, a haploid sperm nucleus and a haploid egg nucleus combine to form the diploid nucleus of the fertilized egg (zygote). The centriole, which is provided by the sperm, will divide to form the two poles of the mitotic spindle during cleavage.

# The prevention of polyspermy

The entrance of multiple sperm **polyspermy** leads to disastrous consequences in most organisms. In the sea urchin, fertilization by two sperm results in a triploid nucleus, in which each chromosome is represented three times rather than twice (Worse condition).

# The prevention of polyspermy

Species have evolved ways to prevent the union of more than two haploid nuclei. The most common way is to prevent the entry of more than one sperm into the egg.

# The prevention of polyspermy

**The sea urchin egg has two mechanisms to avoid polyspermy:**

- A fast reaction, accomplished by an electric change in the egg plasma membrane.
- A slower reaction, caused by the exocytosis of the cortical granules.

# The fast block to polyspermy

# The fast block to polyspermy

The **Fast block to polyspermy** is achieved by changing the electric potential of the egg plasma membrane. This membrane provides a selective barrier between the egg cytoplasm and the outside environment, and the ionic concentration of the egg differs greatly from that of its surroundings.

# The fast block to polyspermy

This concentration difference is especially significant for **sodium and potassium ions.** Seawater has a particularly high sodium ion concentration, whereas the egg cytoplasm contains relatively little sodium. The reverse is the case with **potassium ions.**

# The fast block to polyspermy

This condition is maintained by the plasma membrane, which steadfastly inhibits the entry of sodium ions into the oocyte and prevents potassium ions from leaking out into the environment.

# The fast block to polyspermy

If we insert an electrode into an egg and place a second electrode outside it, we can measure the constant difference in charge across the egg plasma membrane.

# The fast block to polyspermy

This **resting membrane potential** is generally about **70 mV**, usually expressed as 70 mV because the inside of the cell is negatively charged with respect to the exterior.

# The slow block to polyspermy

# The fast block to polyspermy

The eggs of sea urchins (and many other animals) have a second mechanism to ensure that multiple sperm do not enter the egg cytoplasm. **The fast block to polyspermy is transient,** since the membrane potential of the sea urchin egg remains positive for only about a minute.

# The slow block to polyspermy

This brief potential shift is not sufficient to prevent polyspermy, which can still occur if the sperm bound to the vitelline envelope are not somehow removed .

# The slow block to polyspermy

This removal is accomplished by the **cortical granule reaction**, a slower, mechanical block to polyspermy that becomes active about a minute after the first successful sperm-egg attachment.

# The slow block to polyspermy

If we insert an electrode into an egg and place a second electrode outside it, we can measure the constant difference in charge across the egg plasma membrane.

# The slow block to polyspermy

Directly beneath the sea urchin egg plasma membrane are about 15,000 cortical granules. Upon sperm entry, these cortical granules fuse with the egg plasma membrane and release their contents into the space between the plasma membrane and the fibrous mat of vitelline envelope proteins.

# The slow block to polyspermy

These enzymes dissolve the protein posts that connect the vitelline envelope proteins to the cell membrane, and they clip off the bindin receptor and any sperm attached to it.

# The slow block to polyspermy

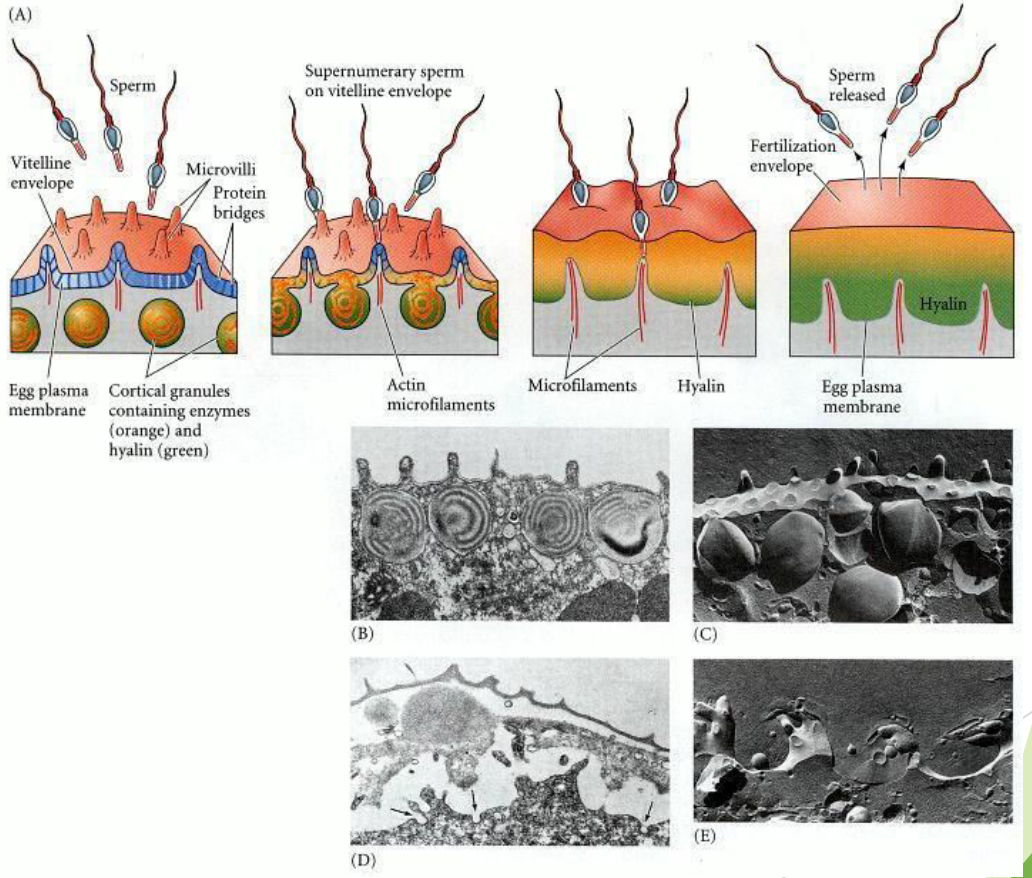
Mucopolysaccharides released by the cortical granules produce an osmotic gradient that causes water to rush into the space between the plasma membrane and the vitelline envelope, causing the envelope to expand and become the **fertilization envelope.**

# The slow block to polyspermy

Mucopolysaccharides released by the cortical granules produce an osmotic gradient that causes water to rush into the space between the plasma membrane and the vitelline envelope, causing the envelope to expand and become the **fertilization envelope.**

# The slow block to polyspermy

A third protein released by the cortical granules, a peroxidase enzyme, hardens the fertilization envelope by crosslinking tyrosine residues on adjacent proteins.



# The slow block to polyspermy

Finally, a fourth cortical granule protein, hyalin, forms a coating around the egg.

The egg extends elongated microvilli whose tips attach to this **hyaline layer**. This layer provides support for the blastomeres during cleavage.

# The slow block to polyspermy

In mammals, the cortical granule reaction does not create a fertilization envelope, but its ultimate effect is the same. Released enzymes modify the zona pellucida sperm receptors such that they can no longer bind sperm.

Effort By: AARISH ALI

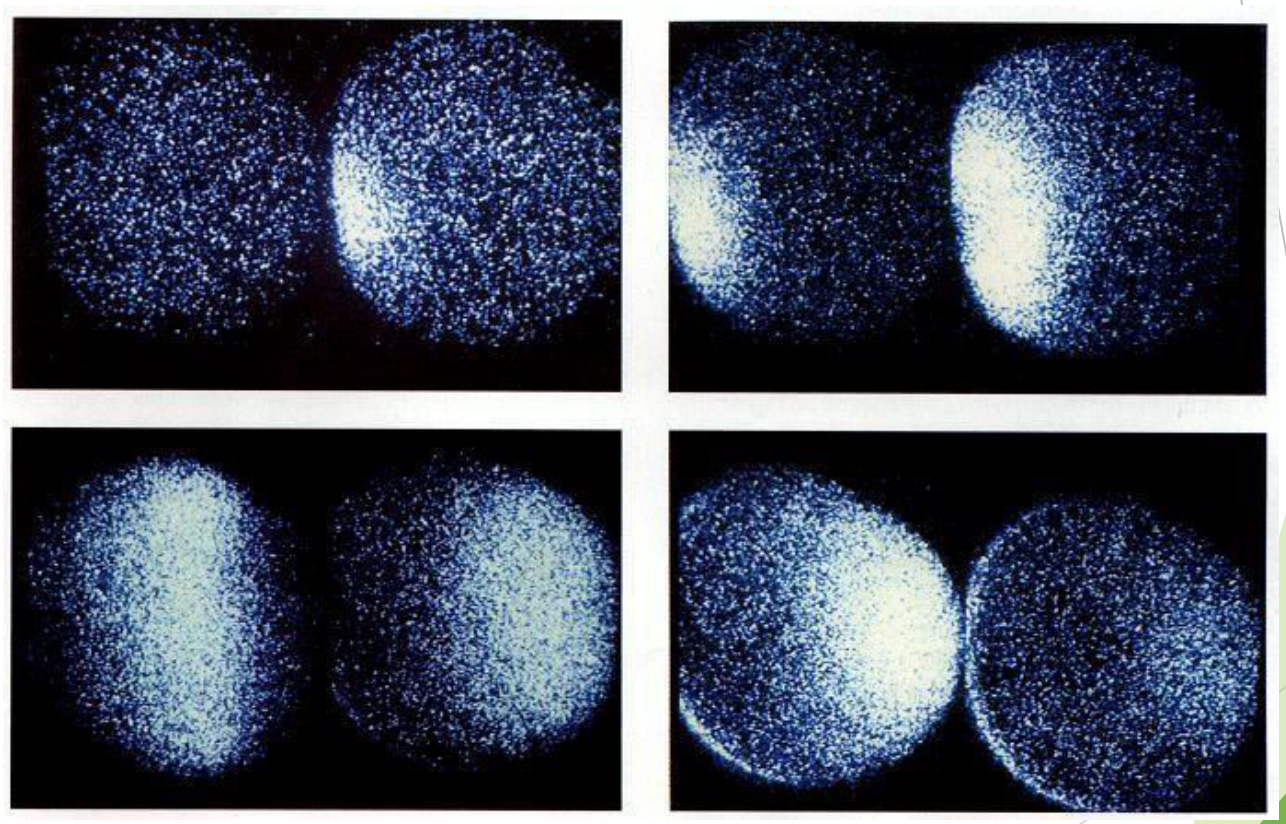
# Role of calcium in cortical reaction

# Role of calcium in cortical reaction

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents.

# Role of calcium in cortical reaction

Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg.



# Role of calcium in cortical reaction

The release of calcium from intracellular storage can be monitored visually using calcium-activated luminescent dyes such as aequorin (isolated from luminescent jellyfish) or fluorescent dyes such as fura-2. These dyes emit light when they bind free calcium ions.

# Role of calcium in cortical reaction

Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

**The drug A23187 is a calcium ionophore** (a compound that transports free calcium ions across lipid membranes, allowing these cations to traverse otherwise impermeable barriers).

# Role of calcium in cortical reaction

Placing unfertilized sea urchin eggs into seawater containing A23187 causes the cortical granule reaction and the elevation of the fertilization envelope.

Therefore, A23187 must be causing the release of calcium ions already sequestered in organelles within the egg.

# Role of calcium in cortical reaction

The calcium ions responsible for the cortical granule reaction are stored in the endoplasmic reticulum of the egg. In sea urchins and frogs, this reticulum is pronounced in the cortex and surrounds the cortical granules.

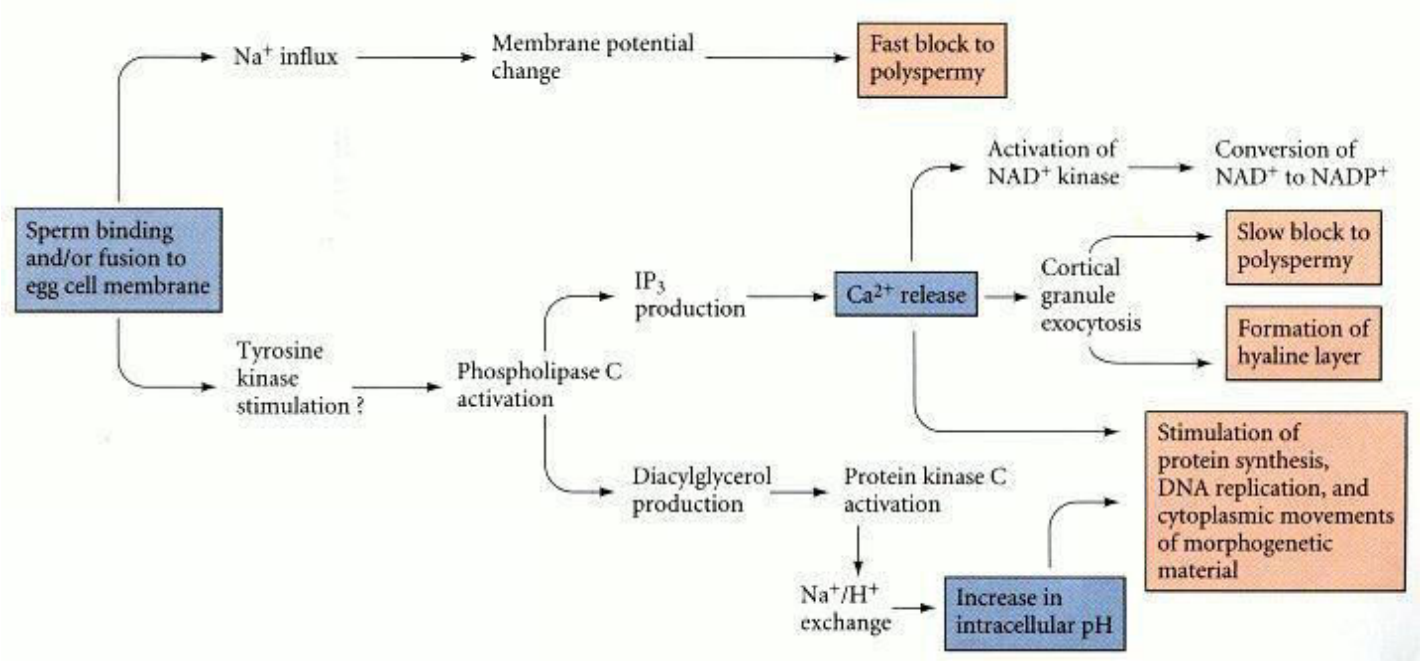
# The Activation of Egg Metabolism

# The Activation of Egg Metabolism

Although fertilization is often depicted as merely the means to merge two haploid nuclei, it has an equally important role in initiating the processes that begin development. These events happen in the cytoplasm and occur without the involvement of the nuclei.

# The Activation of Egg Metabolism

The mature sea urchin egg is a metabolically sluggish cell that is activated by the sperm. This activation is merely a stimulus, however; it sets into action a preprogrammed set of metabolic events.



# The Activation of Egg Metabolism

The responses of the egg to the sperm can be divided into "early" responses, which occur within seconds of the cortical reaction, and "late" responses, which take place several minutes after fertilization begins.

# The Activation of Egg Metabolism

**Early responses:**

Contact between sea urchin sperm and egg activates the two major blocks to polyspermy: the fast block, initiated by sodium influx into the cell, and the slow block, initiated by the intracellular release of calcium ions.

# The Activation of Egg Metabolism

The activation of all eggs appears to depend on an increase in the concentration of free calcium ions within the egg. Such an increase can occur in two ways: calcium ions can enter the egg from outside, or calcium ions can be released from the endoplasmic reticulum within the egg.

# The Activation of Egg Metabolism

**Late responses:**

Shortly after the calcium ion levels rise in a sea urchin egg, its intracellular pH also increases. The rise in intracellular pH begins with a second influx of sodium ions, which causes a 1:1 exchange between sodium ions from the seawater and hydrogen ions from the egg.

# The Activation of Egg Metabolism

This loss of hydrogen ions causes the pH to rise. It is thought that the pH increase and the calcium ion elevation act together to stimulate new protein synthesis and DNA synthesis.

**The late responses of fertilization** brought about by these ionic changes include the activation of DNA synthesis and protein synthesis.

# Introduction to Early Developmental Processes

# Introduction to Early Developmental Processes

## Cleavage

After fertilization, the development of a multicellular organism proceeds by a process called **cleavage**, a series of mitotic divisions whereby the enormous volume of egg cytoplasm is divided into numerous smaller, nucleated cells. These cleavage-stage cells are called **blastomeres**.

# Introduction to Early Developmental Processes

In most species (mammals being the chief exception), the rate of cell division and the placement of the blastomeres with respect to one another is completely under the control of the proteins and mRNAs stored in the oocyte by the mother.

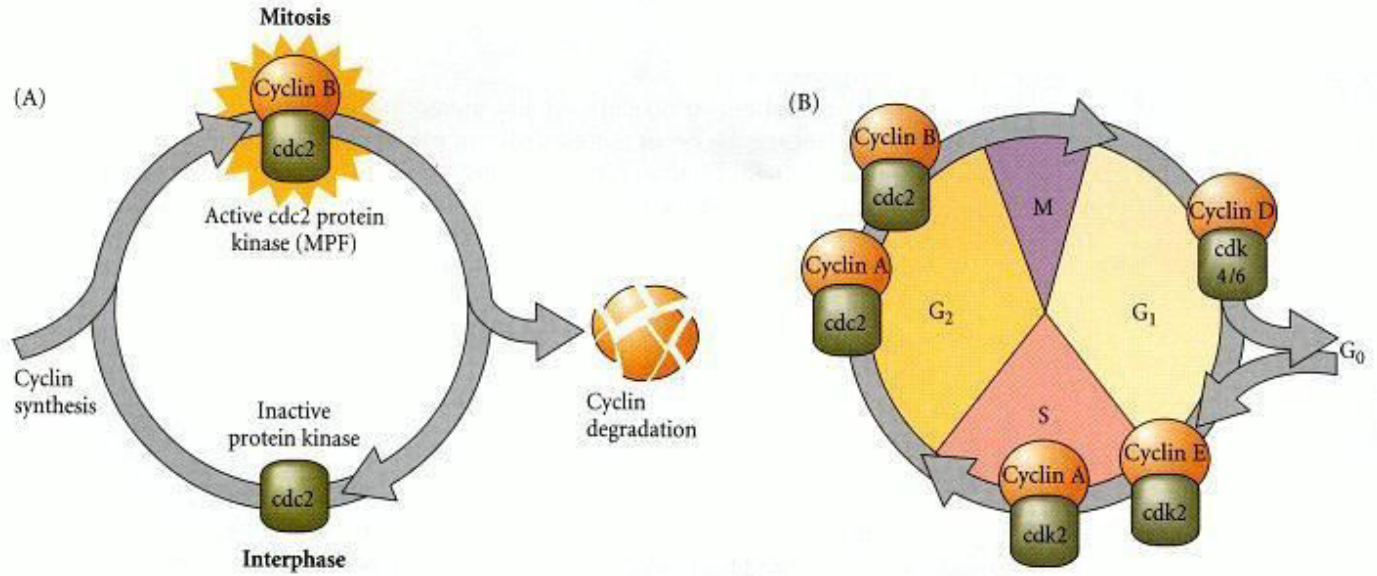
# Introduction to Early Developmental Processes

The transition from fertilization to cleavage is caused by the activation of **mitosis promoting factor** (**MPF**).

MPF was first discovered as the major factor responsible for the resumption of meiotic cell divisions in the ovulated frog egg.

# Introduction to Early Developmental Processes

Blastomeres generally progress through a cell cycle consisting of just two steps: M (mitosis) and S (DNA synthesis).



# Introduction to Early Developmental Processes

Cleaving cells can be experimentally trapped in S phase by incubating them in an inhibitor of protein synthesis. When MPF is microinjected into these cells, they enter M. Their nuclear envelope breaks down and their chromatin condenses into chromosomes. After an hour, MPF is degraded and the chromosomes return to S phase.

# Introduction to Early Developmental Processes

**What causes this cyclic activity of MPF?**

Mitosis-promoting factor contains two subunits.

- The large subunit is called **cyclin B**.
- Small subunit of MPF, the **cyclin-dependent kinase**

# Introduction to Early Developmental Processes

**Cyclin B** is often encoded by mRNAs stored in the oocyte cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the **cyclin-dependent kinase**.

# Introduction to Early Developmental Processes

Cleavage is actually the result of two coordinated processes.

- **karyokinesis**
- **Cytokinesis**

# Introduction to Early Developmental Processes

**karyokinesis** the mitotic division of the nucleus.

The mechanical agent of this division is the mitotic spindle, with its **microtubules** composed of **tubulin** (the same type of protein that makes up the sperm flagellum).

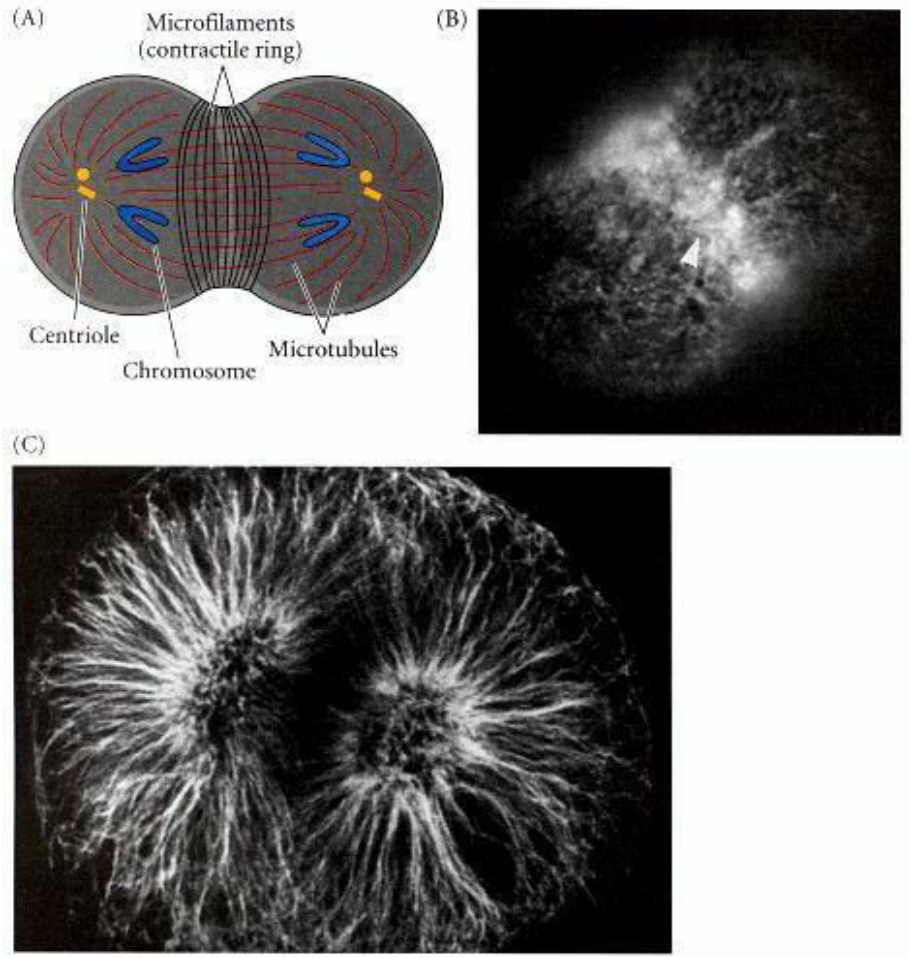
# Introduction to Early Developmental Processes

The second process is **cytokinesis** the division of the cell.

The mechanical agent of cytokinesis is a **contractile ring** of **microfilaments** made of **actin** (the same type of protein that extends the egg microvilli and the sperm acrosomal process).

# Introduction to Early Developmental Processes

The relationship and coordination between these two systems during cleavage is depicted in Figure, in which a sea urchin egg is shown undergoing first cleavage.



# Introduction to Early Developmental Processes

The mitotic spindle and contractile ring are perpendicular to each other, and the spindle is internal to the contractile ring. The contractile ring creates a **cleavage furrow**, which eventually bisects the plane of mitosis, thereby creating two genetically equivalent blastomeres.

# *Patterns of embryonic cleavage*

# *Patterns of embryonic cleavage*

Indeed, different organisms undergo cleavage in distinctly different ways. The pattern of embryonic cleavage particular to a species is determined by two major parameters:

# *Patterns of embryonic cleavage*

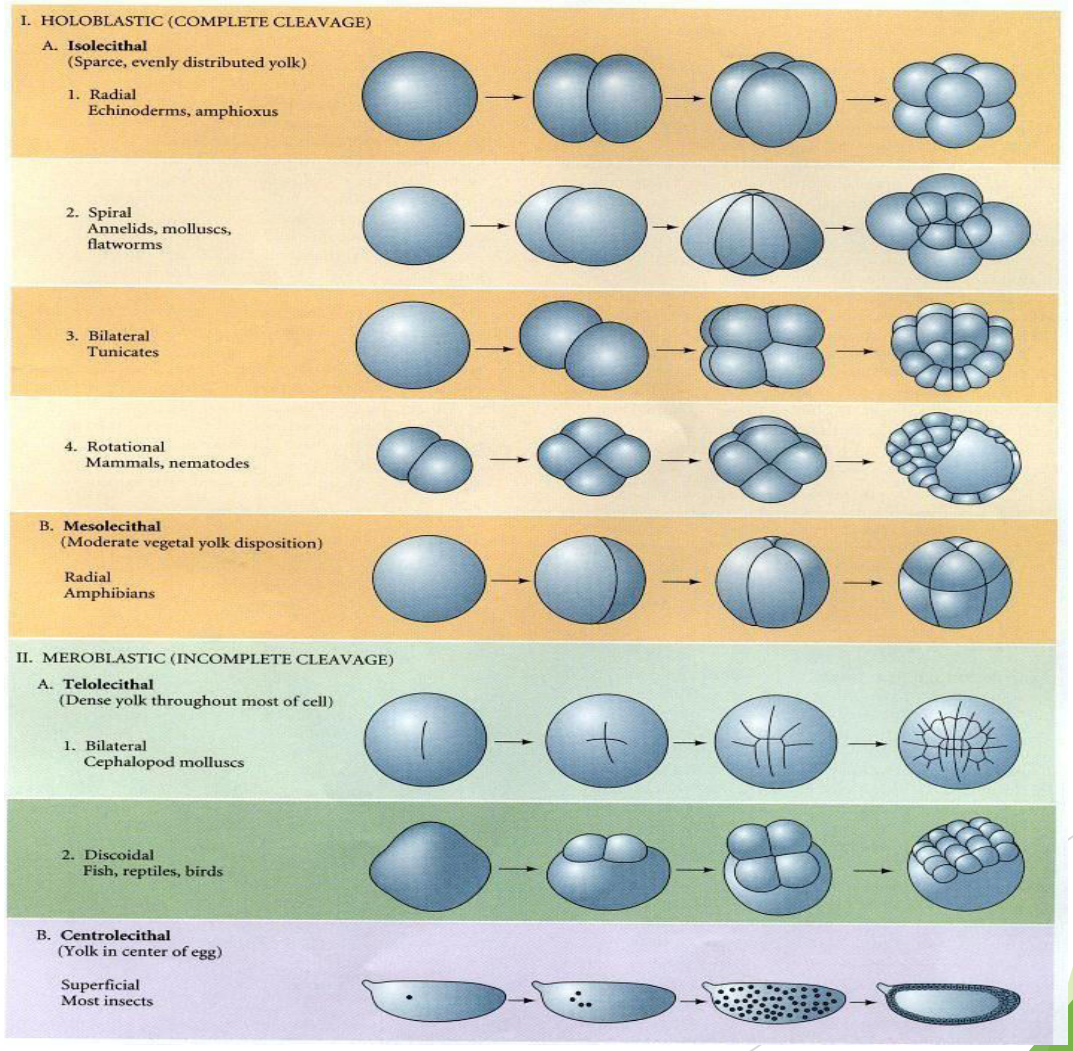
1. The amount and distribution of yolk protein within the cytoplasm.
2. The factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation.

# *Patterns of embryonic cleavage*

The yolk-rich pole is referred to as the **vegetal pole**; the yolk concentration in the **animal pole** is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

# *Patterns of embryonic cleavage*

This diagram provides a classification of cleavage types and shows the influence of yolk on cleavage symmetry and pattern.



# *Patterns of embryonic cleavage*

At one extreme are the eggs of sea urchins, mammals, and snails. These eggs have sparse, equally spaced yolk and are thus **isolecithal** (Greek, "equal yolk"). In these species, cleavage is **holoblastic** (Greek *holos,* "complete"). meaning that the cleavage furrow extends through the entire egg.

# *Patterns of embryonic cleavage*

Zygotes containing large accumulations of yolk undergo **meroblastic** cleavage, wherein only a portion of the cytoplasm is cleaved. The cleavage furrow does not penetrate into the yolky portion of the cytoplasm.

# *Patterns of embryonic cleavage*

The eggs of insects have their yolk in the center (i.e., they are **centrolecithal**), and the divisions of the cytoplasm occur only in the rim of cytoplasm around the periphery of the cell (i.e., **superficial** cleavage).

# *Patterns of embryonic cleavage*

The eggs of birds and fishes have only one small area of the egg that is free of yolk (**telolecithal** eggs), and therefore, the cell divisions occur only in this small disc of cytoplasm, giving rise to the **discoidal** pattern of cleavage.

# **Early responses in** Activation of Egg Metabolism

# **Early responses in** Activation of Egg Metabolism

**Early responses:**

Contact between sea urchin sperm and egg activates the two major blocks to polyspermy: the fast block, initiated by sodium influx into the cell, and the slow block, initiated by the intracellular release of calcium ions.

# Early responses in Activation of Egg Metabolism

The activation of all eggs appears to depend on an increase in the concentration of free calcium ions within the egg. Such an increase can occur in two ways: calcium ions can enter the egg from outside, or calcium ions can be released from the endoplasmic reticulum within the egg.

# Classification of cleavage types

# Classification of cleavage types

At one extreme are the eggs of sea urchins, mammals, and snails. These eggs have sparse, equally spaced yolk and are thus isolecithal (Greek, "equal yolk"). In these species, cleavage is holoblastic (Greek *holos,* "complete"). meaning that the cleavage furrow extends through the entire egg.

# Classification of cleavage types

Zygotes containing large accumulations of yolk undergo meroblastic cleavage, wherein only a portion of the cytoplasm is cleaved. The cleavage furrow does not penetrate into the yolky portion of the cytoplasm.

# Classification of cleavage types

The eggs of insects have their yolk in the center (i.e., they are centrolecithal), and the divisions of the cytoplasm occur only in the rim of cytoplasm around the periphery of the cell (i.e., superficial cleavage).

# Classification of cleavage types

The eggs of birds and fishes have only one small area of the egg that is free of yolk (telolecithal eggs), and therefore, the cell divisions occur only in this small disc of cytoplasm, giving rise to the discoidal pattern of cleavage.

# The Early Development of Sea Urchins

## Cleavage in Sea Urchins

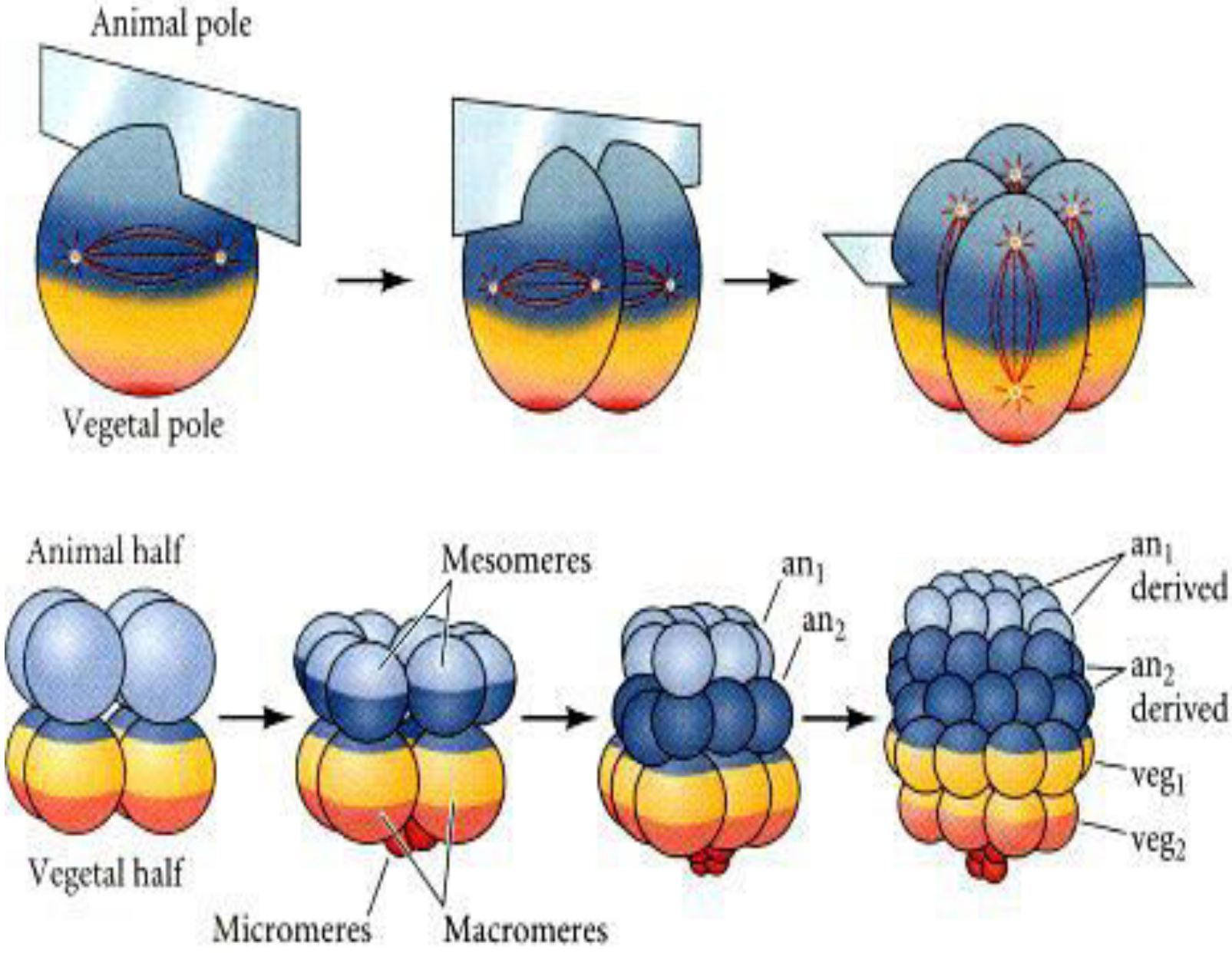
Effort By: AARISH ALI

# Cleavage in Sea Urchins

Sea urchins exhibit radial holoblastic cleavage. The first and second cleavages are both meridional and are perpendicular to each other. That is to say, the cleavage furrows pass through the animal and vegetal poles.

# Cleavage in Sea Urchins

The third cleavage is equatorial, perpendicular to the first two cleavage planes, and separates the animal and vegetal hemispheres from one another. The fourth cleavage, however, is very different from the first three.



# Cleavage in Sea Urchins

The four cells of the animal tier divide meridionally into eight blastomeres, each with the same volume. These cells are called **mesomeres**. The vegetal tier, however, undergoes an unequal equatorial cleavage to produce four large cells, the **macromeres**, and four smaller micromeres at the vegetal pole.

# Blastula formation in Sea Urchan

# Cleavage in Sea Urchins

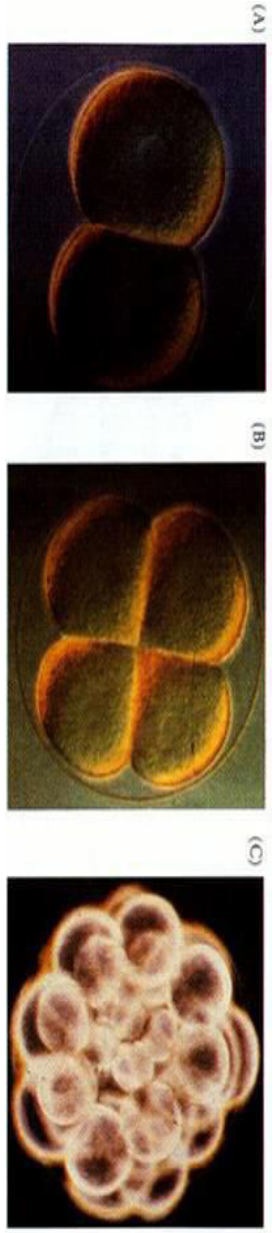
As the 16-cell embryo cleaves, the eight mesomeres divide to produce two "animal" tiers, an1 and an2, one staggered above the other.

The macromeres divide meridionally, forming a tier of eight cells an2.

# Cleavage in Sea Urchins

## Blastula formation

The blastula stage of sea urchin development begins at the 128-cell stage. Here the cells form a hollow sphere surrounding a central cavity, or **blastocoel.**

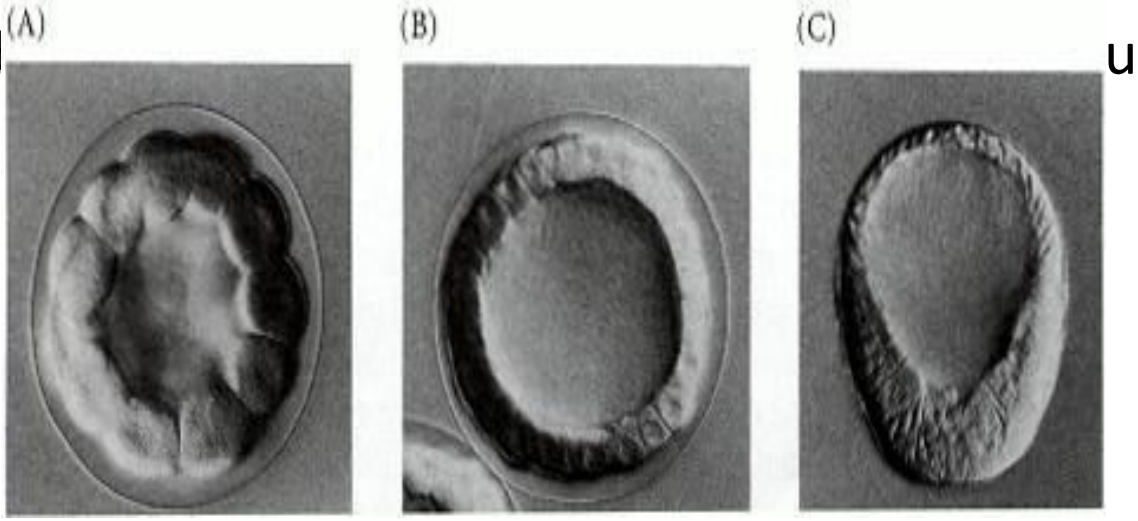


# Cleavage in Sea Urchins

At this time, tight junctions unite the once loosely connected blastomeres into a seamless epithelial sheet that completely encircles the blastocoel. As the cells continue to divide, the blastula remains one cell layer thick, thinning out as it expands.

# Cleavage in Sea Urchins

The cells at the vegetal pole of the blastula begin to thicken, forming a vegetal plate. The cells of the animal half synthesize and secrete a hatching enzyme that digests the fertilization envelope The eml




ula.

# Final Term Syllabus
# Lecture 01 to 72

Efforts By

**AARISH ALI (+96569002641)**

Contact us to join our esteemed group

**“All About Biotech”**

**Note:**

**Students must follow the course announcement for midterm syllabus & any up-gradation in the course.**

REMEMBER IN PRAYERS

KEEP SMILING

SAY NO TO CURRUPTION

HELP TO OTHERS

Regards,

AARISH ALI & TEAM

# “All About Biotech”

+96569002641